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	Departamento Department	Stress Methods & Optimisation	Avión Aircraft	Technologies
<u>Título/Title</u> PROGRAM REMACHADO, USER'S MANUAL				
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<u>Resumen/Summary</u> The purpose of this Technical Report is to define the way of using the REMACHADO program. This program analyses a panel with fastened stiffeners and with a centered crack perpendicular to them. The stiffeners situated parallel to the direction of propagation are considered in the analysis as a rigid support to them. This assumed configuration corresponds to the practical case of a panel stiffened with fastened stringers. This set is, in turn, supported by much more rigid elements, such as ribs or frames. Even though this program is only applicable to symmetric configurations, it can also be used in configurations where the asymmetric arrangements are not significant. This Technical Report supersedes NT-T-ADP-01045 (Program REMACHADO, Version 2.x User's Manual).				
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1 INTRODUCTION

1.1 TECHNICAL SHEET OF THE PROGRAM

Technical Report: TEAS-NT-1710009

<u>PROGRAM</u>	<u>DATE</u>
REMACHADO	2003

OBJECTIVE

Analysis of a panel with axial stiffeners, riveted to the panel in a perpendicular direction to the crack.

<u>LANGUAGE</u>	<u>DEVELOPMENT TOOLS</u>	<u>OS/PLATFORM</u>
Fortran 77	Oracle Solaris Studio	Unix (Solaris) and Windows (x86)
Visual Basic Net	MS Visual Studio 2013	Windows (x86)

<u>NUMBER OF LINES (APPROX.)</u>	<u>EXECUTION MODE</u>
2473 (Fortran, analysis core)	Console (input data in text file)
+ 3240 (Visual Basic .Net)	Interactive GUI
Total 5713	

Notes:

- Blank lines and commented lines excluded
- Count for Visual Basic code excludes lines created by Visual Studio in designer.vb files

<u>AUTHOR</u>	<u>VERSION</u>
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ADDITIONAL DOCUMENTS

See list of applicable references in chapter §1.3.

1.2 SOFTWARE PURPOSE AND DESCRIPTION

This program analyses a panel with fastened stiffeners (stringers) and with a centered crack perpendicular to them. The stiffeners situated parallel to the direction of propagation are considered in the analysis as a rigid support to the stringers.

The analysis is performed under the following hypotheses:

- The skin panel is supposed to be infinite.
- The crack is assumed to be symmetric with respect to the crack propagation direction and with respect to its axis.
- The crack growing section coincides with the central transversal support (frame/rib).
- The stiffeners are able to carry axial and bending load. For the crack propagation analysis, the stiffeners are supposed to be supported on the transversal supports, which are parallel to the crack propagation direction. The bending stiffness of the transversal supports is assumed to be infinite.
- The stiffeners initial state and failure criterion is specified in the input data file. They can be considered as intact or broken.

Figure 1 shows the crack configuration supposed in the analysis. This configuration corresponds with a skin panel with a crack growing in the direction of the frames or the ribs, depending on the structural element considered. The X axis is always coincident with the crack growing direction.

For other configurations such as a crack growing parallel to the stringers or in a skin section between two frames/ribs, please refer to the considerations outlined in section 5.

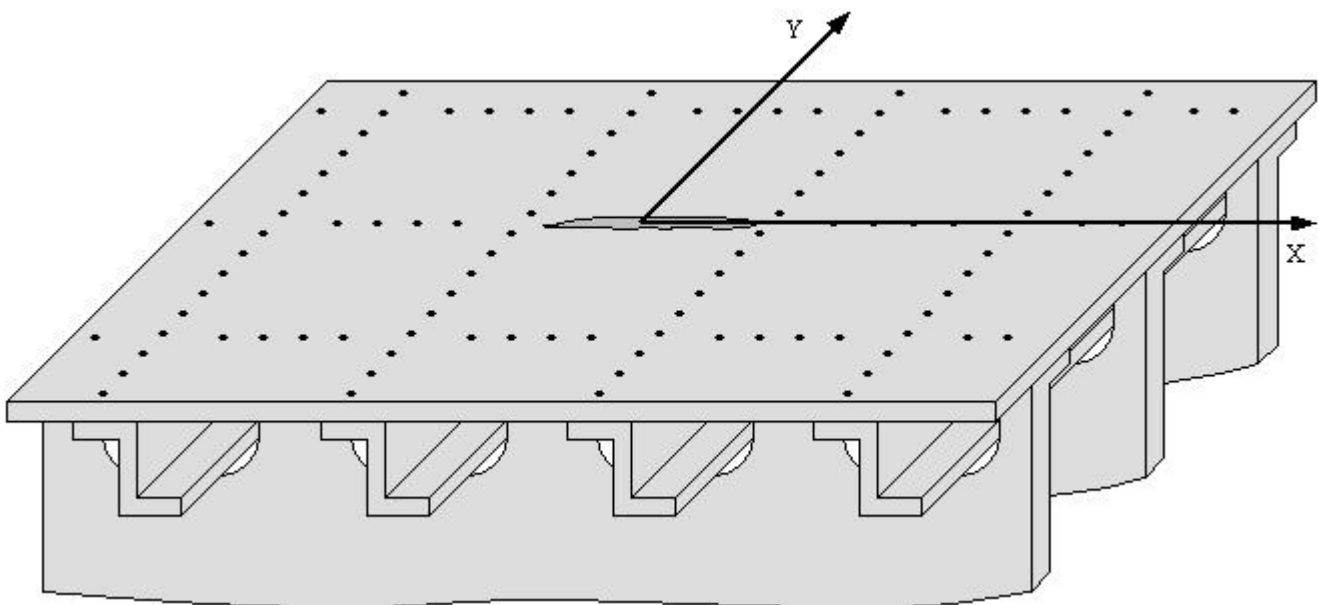


Figure 1. Crack configuration supposed in the analysis

REMACHADO provides the following results for each crack length specified in the input file:

- Stress intensity factor and normalised stress intensity factor (factor β).
The β factor obtained from REMACHADO can be used in a crack propagation analysis (i.e. as beta 35/37 in GRIETA) in order to consider the effect of the stiffeners in the crack growth.
- Stress field in the stiffener.
- Residual strength of the skin.
- Load transferred by the fasteners.

1.3 REFERENCES

	<i>Document Reference</i>	<i>Issue</i>	<i>Title</i>
Ref. 1	NT-T-ADP-03002	A	Theoretical Manual of the REMACHADO Program.
Ref. 2	NT-T-ADP-02010	A	Programmer Manual of the REMACHADO Program.
Ref. 3	ME-T-NT-130063	D	Manual de usuario del programa UTILIDADES.

1.4 DIFFERENCES AMONG VERSIONS

Version 1.0:

First issue. Initial issue developed in FORTRAN 77 for VAX computer.

Version 2.0:

Program adapted to Sun Solaris computer.

Version 2.1:

The language of the dialogs and output file can be chosen by the user between Spanish or English.

The source code is restructured in order to allow the creation of a DLL for Windows OS.

A Windows application has been developed as GUI. This application uses the same input file and provides the same output results. The analysis is performed by calling to the DLL, compiled from the same source code as the official FORTRAN version.

Version 2.2:

An error in the equivalent thickness calculation (skin + stiffener) has been corrected.

Version 3.0:

The GUI has been improved.

The program warns if no solution is found for the system of equations for calculating the fasteners forces.

Program adapted to the new development environment:

- Visual Studio 2010.
- Intel Parallel Studio XE 2013.

Version 3.1:

Uncontrolled bugs in the GUI have been fixed.

Version 3.2:

The error handling has been improved. The execution is aborted when the input data is not consistent and a text message with a brief description of the error is shown.

The accuracy of the calculation has been increased.

2 PROGRAM EXECUTION

2.1 WINDOWS/PC VERSIONS

The Visual Basic executable program REMACHADO.exe (as well as an electronic copy of this manual and the program examples) is located in the folder:

```
\\Cae\InfoCalculo\Soft\remachado
```

To execute the program, the user has to click twice on the Visual Basic executable program.

There is another executable file with the same name, which is the Fortran program in console mode, in the subdirectory .Consola within that folder (and another copy of the console-mode executable in \\Cae\InfoCalculo\Soft\Consola, this last path than can be added to the Windows system PATH to facilitate its use).

2.2 UNIX/MULTIVAC VERSION

To run interactively this version of the program (console application), you must have already done the assignment of the symbols list in Airbus DS Structural Analysis Office using the command:

```
. asignaciones_calculo
```

This instruction can be executed in the commands terminal prior to execute the program, but it is strongly recommended to include this statement in the files ".profile", and ".kshrc" (if working in Korn Shell) or ".bashrc" (if you work in Bash Shell). To do this, the following lines should be included in the file ".profile":

```
. asignaciones_calculo
ENV=$HOME/.kshrc
export ENV
BASH_ENV=$HOME/.bashrc
export BASH_ENV
```

In files ".kshrc" and ".bashrc" the following line shall be added:

```
. asignaciones_calculo
```

The instruction to run the program in interactive mode in the MULTIVAC server is:

```
$remachado
```

The program asks for the language to use and for the name of the input and output files:

```
Idioma / Language:
```

```
=====
```

```
1 ... ESPAÑOL
```

```
2 ... ENGLISH
```

```
Idioma / Language: 2
```

```
Name of the input file: Example1.dat
```

```
Name of the output file: Example1.out
```

This console application can be executed in one single command, by passing as arguments the input data file, the desired name of the output results file, and an integer number to select the report language (1 is English, 2 is Spanish):

```
$remachado filein.dat fileout.txt 2
```

If the third parameter (language) is omitted, English language will be used.

2.3 LINUX/MULTIVAC VERSION

To run interactively this version of the program (console application) from the Multivac3 server (Unix Solaris), you must have already done the assignment of the symbols list in Airbus DS Structural Analysis Office (see previous chapter).

The Linux version of this tool can be run from the Unix Solaris command line terminal in the Multivac3 server, even the Linux HPC cores will do the work in the background (a special wrapper acts as interface between these platforms). The instruction to run the Linux version of the program in interactive or single-command modes in the MULTIVAC3 server is:

```
remachado
```

```
remachado filein.dat fileout.txt 2
```

3 INPUT DATA

Figure 2 shows the initial crack configuration assumed by the program REMACHADO and the main geometrical parameters:

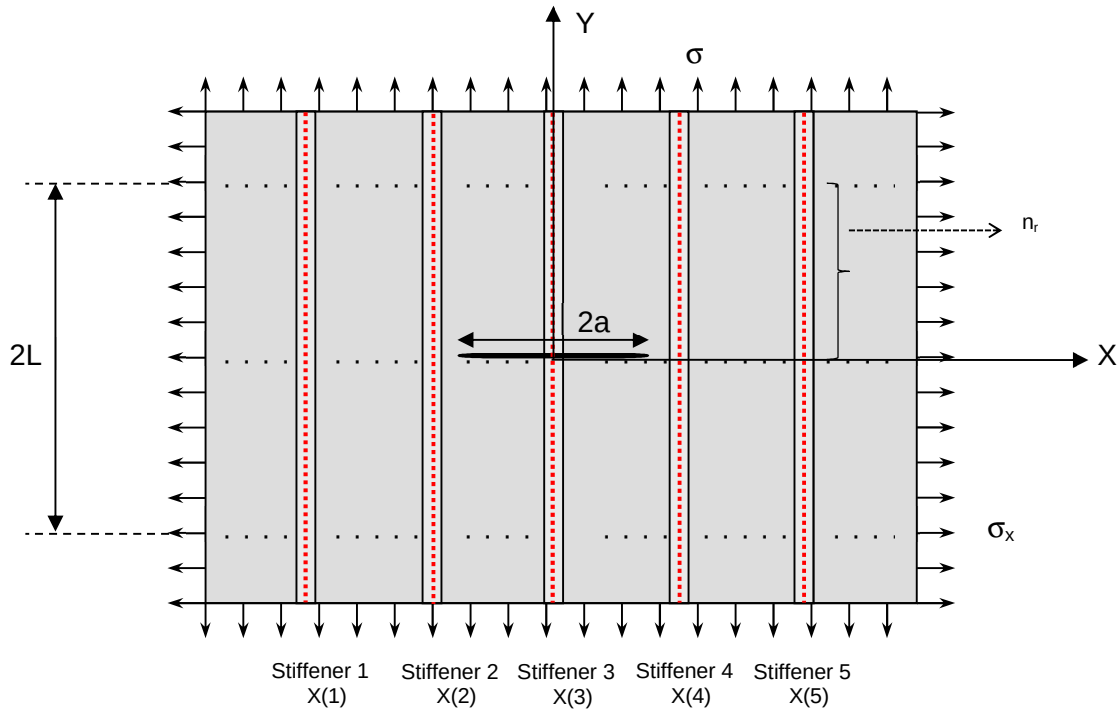


Figure 2. *Initial Crack Configuration*

- L: Distance between transversal supports; ribs or frames, depending on the structural component to be considered.
- a: Semi-crack length.
- σ : Axial stress of reference in the far field.
- σ_x : Transversal stress of reference in the far field.
- n_r : Number of fasteners between transversal supports.
- $X(i)$: X coordinate of the stringer i .

It is important to note that the X axis is always coincident with the crack growing direction.

For other configurations such as a crack growing parallel to the stringers or in a skin section between two frames/ribs, please refer to the considerations outlined in section 5.

3.1 WINDOWS PC/GUI

The input data has to be typed in the application form or provided in an input file.

Figure 3 shows the appearance of the application form after the loading of the example 1 input file.

Remachado - C:\Users\C83801\Documents\Wip\Ejemplo1.dat

Run 1 Run 2

☒ Title Program REMACHADO, Example 1.
Crack with central stiffener intact.
Material: Skin: 2024 T3. Stiffeners: 7075 T73511.
Uniform fastening, 15 fasteners per stringer.

☒ Print control
☒ Results for the skin
☒ Results for stringers
☒ Allowable stresses
☒ Fastener loads

☒ Panel data
E 72400.00 Mu 0.33 t 2.00 Kc 5000.00 L 500.00

☒ Far field stress
Saxial 200.00 Slat/Sax 0.50

☒ Stringer data

	X	t	Ag	An	boxg	boxn	Zo	Zi	E	Ftu	Fty	Failure
1	0.00	3.00	237.00	222.60	44610.00	40674.00	17.50	17.50	71700.00	469.00	400.00	0: Normal
2	100.00	3.00	237.00	222.60	44610.00	40674.00	17.50	17.50	71700.00	469.00	400.00	0: Normal
3	200.00	3.00	237.00	222.60	44610.00	40674.00	17.50	17.50	71700.00	469.00	400.00	0: Normal

No. of stringers 3

☒ Fastener data

Fastening type Uniform

	Num	Pitch	Diameter	Er	Stiffness
Stringer 1	15	20.00	4.80	71000.00	0.00

☒ Crack lengths

No. of lengths 21

	Length
1	0.00
2	10.00
3	20.00
4	30.00
5	40.00

Output file C:\Users\C83801\Documents\Wip\Ejemplo1.txt

Input data:

Language: ☐ Español ☒ English

«New run»

«Remove run»

Program execution

Figure 3. Input data form.

Since this example includes two problems to be solved, the form has two tabs, one per problem, labelled "Run 1" and "Run 2", as shown above.

The form shows the data corresponding to the selected tab ("Run 1" in the above figure). The data corresponding to the second problem can be seen clicking on the tab "Run 2".

The program can be executed clicking the button "Program execution".

3.2 INPUT FILE

The input data file contains all the necessary data that define the problem, as well as the different options required for solving it.

Basically, it contains eight data groups, all of which are necessary. They will be numbered from 1 to 8 and should appear in strict order, from lower to higher.

Each set of 8 groups makes up the solution to a separate problem. If we want the solution for different problems, they must appear in the input file, as many sets as well as number of different problems.

These sets will be included one after the other. Each set of problem data will contain only the data groups that are different from the previous one.

All the numerical data contained in this file will be read in a free format. The variables will be separated by one or more blank spaces. The numeration of each group will be included in the first column.

Listed below are the contents of each data group:

GROUP 1: Title of the problem.

Four lines making up the title will be included (66 characters at most), which will appear at the heading of each page of the output file:

```
Line 1 -----> 1 First line of the title
Line 2 ----->   Second line of the title
Line 3 ----->   Third line of the title
Line 4 ----->   Fourth line of the title
```

GROUP 2: Data of the skin.

The structure of the data of this group is the following:

```
Line 1 ----->      2 Optional comment
Line 2 ----->      E  μ  t  Kc  L
```

Where: E = Young modulus of the skin.

μ = Poisson modulus

t = Panel thickness

Kc = Fracture toughness.

L = Distance between transversal supports (see Figure 2). If there is no transversal supports (infinite spacing), the parameter L has to be defined as 0. (See section 5 for further details).

GROUP 3: Data on the stiffeners.

The structure of the data of this group is the following:

Line 1 ---->	3	Optional comment			
Line 2 ----->	nl	X(l)	X(2)	...	X(nl)
Line 3 ----->		ts(l)	ts(2)	...	ts(nl)
Line 4 ---->		Ag(1)	Ag(2)	...	Ag(nl)
Line 5 ---->		An(1)	An(2)	...	An(nl)
Line 6 ---->		Ig(1)	Ig(2)	...	Ig(nl)
Line 7 ---->		In(1)	In(2)	...	In(nl)
Line 8 ---->		Zo(1)	Zo(2)	...	Zo(nl)
Line 9 ---->		Zi(1)	Zi(2)	...	Zi(nl)
Line 10 ---->		Es(1)	Es(2)	...	Es(nl)
Line 11 ---->		Ftu(1)	Ftu(2)	...	Ftu(nl)
Line 12 ---->		Fty(1)	Fty(2)	...	Fty(nl)
Line 13 ---->		Fail(1)	Fail(2)	...	Fail(nl)

Where: nl = Number of axial stiffeners.

X(i) = Coordinate X of each stiffener. It must be ≥ 0 .

ts(i) = Thickness of the flange of the stiffener fastened to the skin.

Ag(i) = Gross area (area of the stiffener without considering the existence of holes).

An(i) = Net area (area of the stiffener considering the existence of holes).

Ig(i) = Gross moment of inertia (moment of inertia of the stiffener without considering the existence of holes).

In(i) = Net moment of inertia (moment of inertia of the stiffener considering the existence of holes).

Zo(i) = Distance from the centre of gravity to the contact surface with the skin (see Figure 4).

Zi(i) = Distance from the centre of gravity to the inner flange (see Figure 4).

Es(i) = Young's modulus.

Ftu(i) = Ultimate tensile stress of the material.

$F_{ty}(i)$ = Yield tensile stress of the material.

$Fail(i)$ = Initial state and failure criteria.

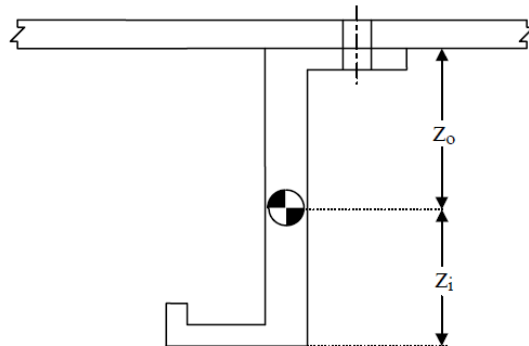


Figure 4. Stiffener - Z_o and Z_i definition

The stiffener position should be defined in such a way that their x coordinate is arranged from the least to the greatest.

The parameter $Fail(i)$, which defines the initial state and the failure criteria of the stiffener "i", has the following meaning, according to its value:

$Fail(i) = -1$ The stiffener is initially intact and will never fail.

$Fail(i) = 0$ The stiffener is initially intact, but could break in accordance with the criteria established in the theoretical manual.

$Fail(i) = 1$ The stiffener is broken from the beginning.

GROUP 4: Data on the riveting of each stiffener.

Two types of riveting can be defined, either uniform or variable.

The case of UNIFORM fastening is that in which all the fasteners of a stiffener are equals, they are aligned along the line defined by coordinate X of the stiffener, and the step between the fasteners is constant. In this case, the structure of this Group would be the following:

Line 1 -----> 4 UNI Optional comment

[nl+1] lines -----> nr(1) Pitch(1) Dm(1) Erm(1) Stm(1)

nr(2) Pitch (2) Dm(2) Erm(2) Stm(2)

.....

.....

nr(nl) Pitch (nl) Dm(nl) Erm(nl) Stm(nl)

A VARIABLE fastening is the one in which the riveting of a stiffener ($_*$ of all) does not comply with some of the conditions imposed for the uniform fastening. In this case, the structure of the group would be the following:

```

Line 1      ----->  4  VAR Optional comment

[nr(1)] lines ----->      nr(1)  Xr(1,1)  Yr(1,1)  D(1,1)  Er(1,1)  St(1,1)
                                Xr(1,2)  Yr(1,2)  D(1,2)  Er(1,2)  St(1,2)
                                .....
                                .....

                                Xr(1,nr) Yr(1,nr) D(1,nr) Er(1,nr) St(1,nr)
[nr(2)] lines ----->      nr(2)  Xr(2,1)  Yr(2,1)  D(2,1)  Er(2,1)  St(2,1)
                                Xr(2,2)  Yr(2,2)  D(2,2)  Er(2,2)  St(2,2)
                                .....
                                .....

                                Xr(2,nr) Yr(2,nr) D(2,nr) Er(2,nr) St(2,nr)
[nr(2)] lines ----->      nr(2)  Xr(2,1)  Yr(2,1)  D(2,1)  Er(2,1)  St(2,1)
                                .....
                                .....

```

And so successively, with as many sets of lines as stiffeners.

The meaning of each of the prior magnitudes, defined for the case of a uniform fastening, is the following:

- nr(i) = Number of fasteners of the stiffener "i". See Figure 2.
- Pitch(i) = Spacing of the fasteners of the stiffener.
- Dm(i) = Diameter of the fasteners.
- Erm(i) = Young's modulus of the fasteners.
- Stm(i) = Stiffness of the fasteners. Stm = 0 for automatic calculation.

In the case of a variable fastening, the significance of each magnitude is:

- Nr(i) = Number of fasteners of the stiffener "i". See Figure 2.
- Xr(i, j) = Coordinate X of the fastener "j" of the stiffener "i". It must be ≥ 0 .
- Yr(i, j) = Coordinate Y of the fastener. It must be ≥ 0 .
- D(i, j) = Diameter of the fastener.
- Er(i, j) = Young's modulus of the fastener.
- St(i, j) = Stiffness of the rivet. St = 0 for automatic calculation.

In both cases of fastening, uniform and variable, if the stiffness of the fasteners is defined (St or Stm), it will be the value which the program will consider in the calculations.

The fasteners position should be defined in such a way that their y coordinate is arranged from the least to the greatest.

If a null value is defined for the stiffness, the program will calculate it in accordance with the following expression (see Ref. 1):

$$St = \frac{\left[\frac{2 \cdot D}{t + t_s} \right]}{2.2 \cdot \left[\frac{1}{t \cdot E} + \frac{1}{t_s \cdot E_s} + \frac{1}{2 \cdot t \cdot E_r} + \frac{1}{2 \cdot t_s \cdot E_r} \right]}$$

- Where:
- D = Diameter of the rivet.
 - E = Young's modulus of the skin.
 - E_s = Young's modulus of the stiffener.
 - E_r = Young's modulus of the fastener.
 - t = Thickness of the skin.
 - t_s = Thickness of the stiffener in the attachment area.

GROUP 5: Stresses in the far field.

The structure of the data of this group is the following:

Line 1 -----> 5 Optional comment

Line 2 -----> σ R

Where: σ = Axial stress of reference in the far field. This stress is defined as the axial load applied to the set of panels and stiffeners, divided by their total area (see Figure 2 and Figure 5).

R = Ratio between the transversal stress σ_x and the reference stress σ .

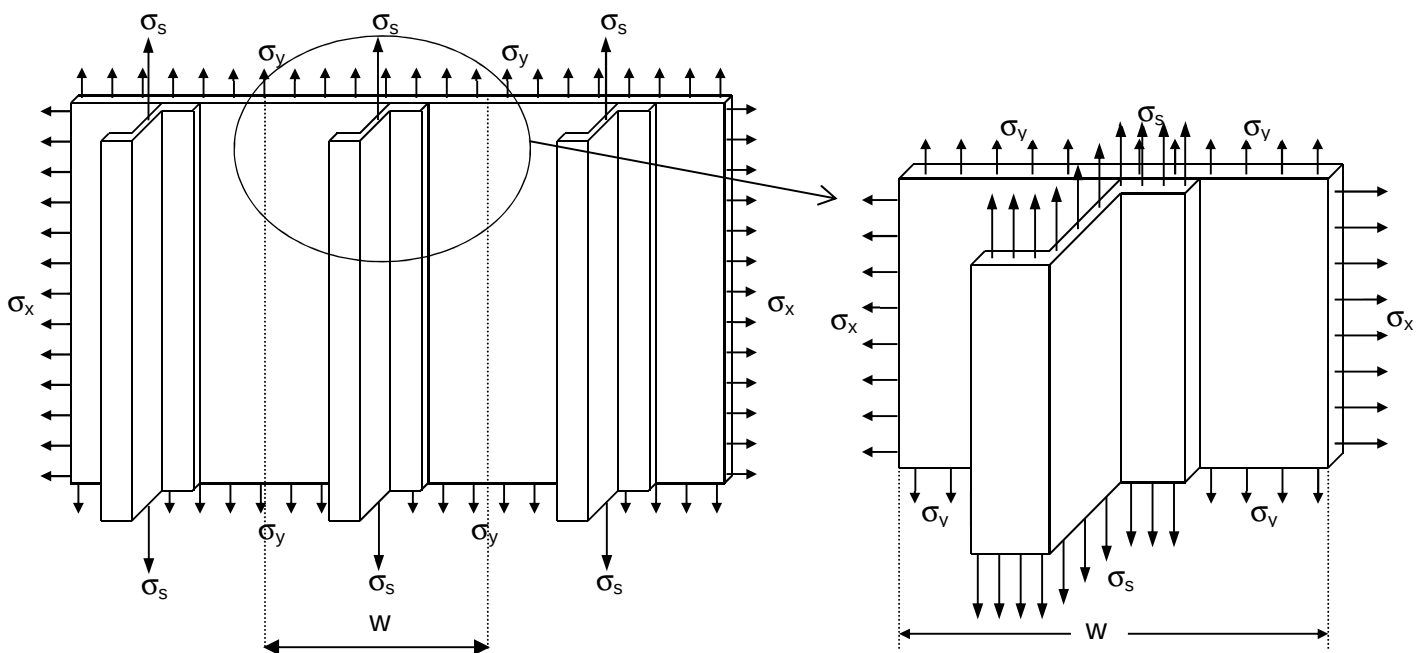


Figure 5. Stress field in the skin and in the stiffener

The reference stress, σ , has to be calculated as:

$$\sigma = \frac{\sigma_s \cdot A + \sigma_y \cdot t \cdot w}{A + t \cdot w}$$

Where: σ_s = Stress in the stiffener

$$\frac{\sigma_y - \mu \cdot \sigma_x}{E} = \frac{\sigma_s}{E_y}$$

σ_y = Stress in the skin in the Y direction

A = Stiffener area

w = Distance between stiffeners

t = Skin thickness

GROUP 6: Crack lengths.

The structure of the data of this group is the following:

Line 1 -----> 6 Optional comment.

Line 2 -----> na a(1) a(2) ... a(na)

Where: na = Number of crack lengths to be analysed.

a(i) = Each one of the "na" crack lengths (see Figure 2).

It is not necessary for the crack lengths to be defined in an orderly manner, as the program will put them in order automatically from smaller to larger.

GROUP 7: Print control.

The structure of the data of this group is the following:

Line 1 -----> 7 Optional comment.

Line 2 -----> I(1) I(2) I(3) I(4)

Where: I(1) = Printing code for skin results.

I(2) = Printing code for stiffeners results.

I(3) = Printing code for allowable stresses and residual strength results.

I(4) = Printing code for fastener loads.

These codes are defined as follows:

I(i) = 0: The corresponding results are not written in the output file.

I(i) = 1: The corresponding results are written in the output file.

GROUP 8: Execution of the program.

This group invokes the execution of the program and should appear as many times as problems there are to resolve.

The structure of the data in this group is the following:

Line 1 -----> 8 Optional comment

4 OUTPUT RESULTS

4.1 WINDOWS PC/GUI

The program is executed clicking the button "Program execution". Then the application starts to solve the problems showing the following form:

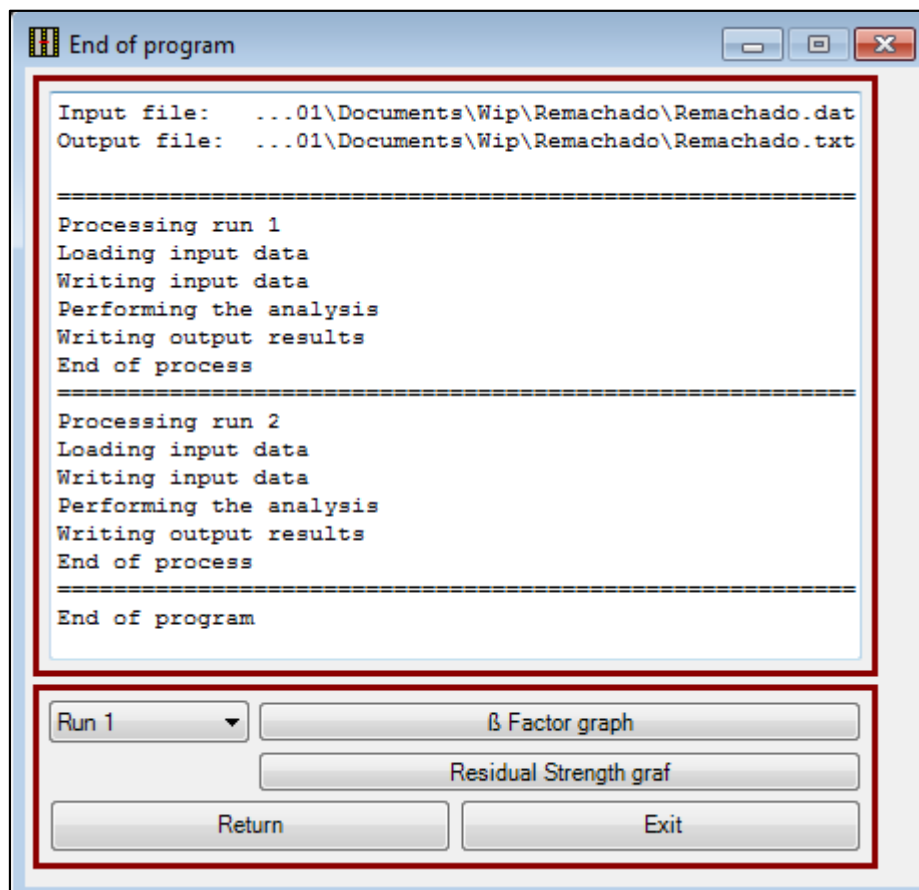


Figure 6. *Tracking and results form.*

The output results are written in the output file, but it is possible to see some of them in graphic way. For this purpose, there are two buttons to show the graphic of beta factor or residual strength. These buttons are only visible when the program execution has finished.

The graphics corresponding to the first problem of the above example (Example 1, detailed in section 5) are shown below:

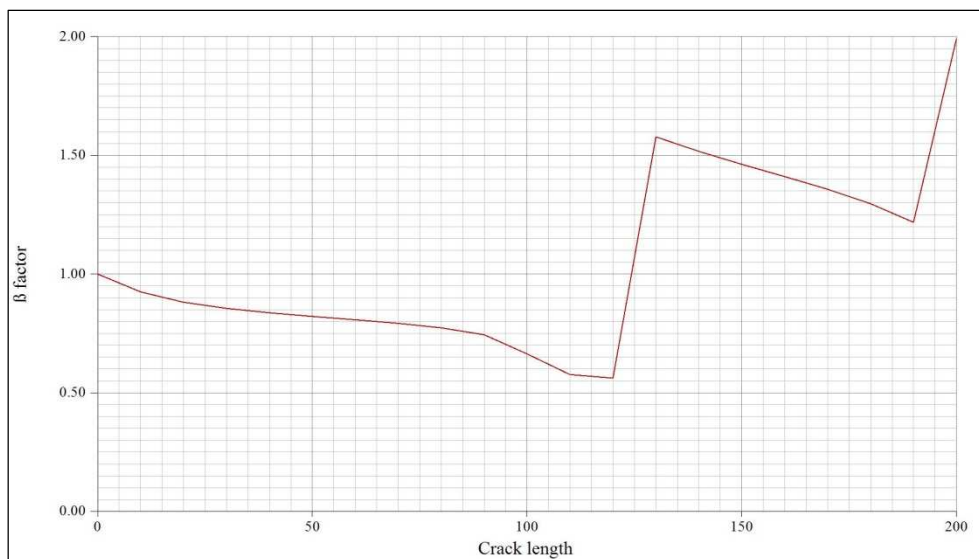


Figure 7. *Beta factor graph.*

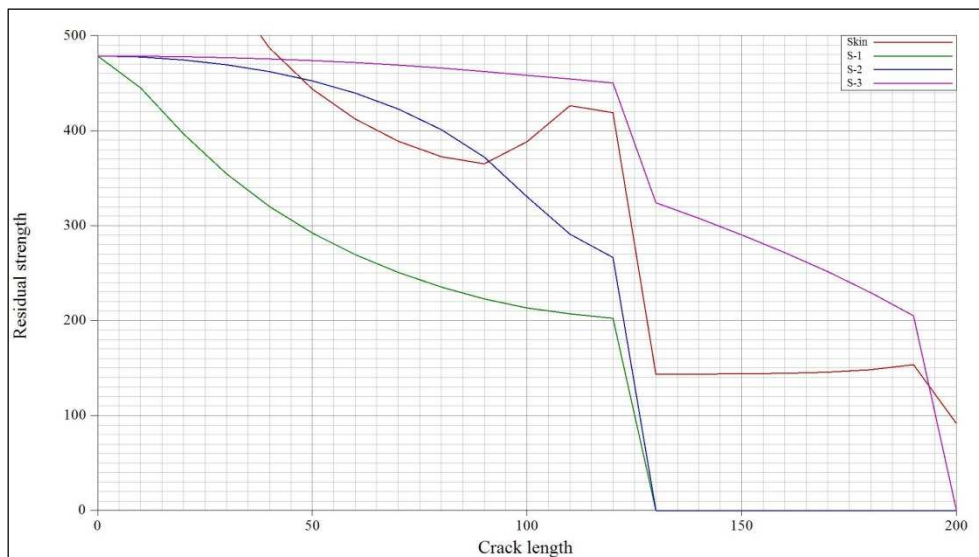


Figure 8. *Residual strength graph.*

4.2 OUTPUT FILE

The REMACHADO program creates a single output file. This file contains the results of the analysis for the different configurations and crack lengths, which have been specified in the input file.

This file presents the following information for each one of the analyses of the program:

- Summary of the input data:
 - Data of the skin.
 - Data of the stiffeners.
 - Data of the fasteners.
 - Data of the far field stress.
- Results for the skin. For each crack length:
 - Stress distribution in the far field (for $X = a$)
 - Stress intensity factor.
 - Factor β (normalised stress intensity factor).
 - State of each stiffener (Intact or broken).
- Results for the stiffeners. For each crack length:
 - Stress in the far field (regardless of the length of the crack).
 - Axial load.
 - Bending moment.
 - Gross stress on the extreme lines of the stiffener.
 - Net stress on the extreme lines of the stiffener.
- Allowable stresses. For each crack length:
 - Residual strength of the skin.
 - Allowable stress for each stiffener according to the criteria of F_{tu} .
 - Allowable stress for each stiffener according to the criteria of F_{ty} .
- Load transferred by the fasteners for each crack length.
- Errors detected in the input data.

In Chapter 6, two examples are given of the output file.

4.3 WARNING AND ERROR MESSAGES

When the program detects an error in the input data, it writes it on the results file and interrupts the running of the program. The error messages, which might appear, are detailed below:

Error messages in the input data. English output:

ERROR: The input file cannot be opened.
ERROR: Erroneous number of data group in the input file.
ERROR: Too many stiffeners exceeding the program capability.
ERROR: Too many crack lengths for analysis, exceeding the program capability.
ERROR: Too many fasteners for a stiffener exceeding the program capability
ERROR: Group 1 missing in the input file (Title).
ERROR: End of file while reading group 1 in the input file (Title).
ERROR: Group 2 missing in the input file (Skin).
ERROR: End of file while reading group 2 in the input file (Skin).
ERROR: The definition of Group 2 is not correct (Skin).
ERROR: Group 3 missing in the input file (stiffener).
ERROR: End of file while reading group 3 in the input file (Stiffener).
ERROR: The definition of Group 3 is not correct (Stiffener).
ERROR: Group 4 missing in the input file (Fasteners).
ERROR: End of file while reading group 4 in the input file (Fasteners).
ERROR: The definition of Group 4 is not correct (Fasteners).
ERROR: Group 4 (Fasteners) cannot be defined before group 3 (Stiffener).
ERROR: Unknown type of fasteners distribution.
ERROR: Group 5 missing in the input file (Stress field).
ERROR: End of file while reading group 5 in the input file (Stress field).
ERROR: The definition of Group 5 is not correct (Stress field).
ERROR: Group 6 missing in the input file (Crack lengths).
ERROR: End of file while reading group 6 in the input file (Crack lengths).
ERROR: The definition of Group 6 is not correct (Crack lengths).
ERROR: Group 7 missing in the input file (Print control).
ERROR: End of file while reading group 7 in the input file (Print control).
ERROR: The definition of Group 7 is not correct (Print control).
ERROR: The output file cannot be opened.
ERROR: The output file cannot be written.
ERROR: The definition of the skin panel material properties is not correct. The value of the material properties must be greater than 0.
ERROR: The definition of the skin panel geometry is not correct. The panel thickness must be greater than 0.
ERROR: The definition of the stiffener material properties is not correct. The value of the material properties must be greater than 0.
ERROR: Error in the definition of the ultimate and yield tensile strength of the stiffener material (Ftu<Fty).
ERROR: The definition of the stiffener geometry is not correct. All the geometry properties must be greater than 0.
ERROR: The definition of the stiffener geometry is not correct (Agross<Anet, Igross<Inet).
ERROR: The definition of the riveting material properties is not correct.
ERROR: The fastener pitch is not correct (pitch<diameter).
ERROR: The definition of the riveting geometry is not correct. The values of the properties must be greater than 0.

- ERROR: The definition of the stiffener`s position is not correct. The stiffener position should be defined in such a way that their X coordinate is arranged from the least to the greatest.
- ERROR: The distance between stiffeners is lower than the rivet diameter.
- ERROR: The definition of the fasteners position is not correct. The fasteners to be considered in the analysis cannot be located beyond the first transversal support (L).
- ERROR: The X coordinate of some of the fasteners is not consistent with the X coordinate of the stiffeners.
- ERROR: The definition of the panel geometry is not correct. The distance between transversal supports (L) cannot be lower than 0.
- ERROR: The reference stress is not correct. The reference stress value must be greater than 0.'
- ERROR: The definition of the stiffeners position is not correct. The X coordinate of the axial stiffeners must be greater than 0.
- ERROR: The definition of the crack lengths is not correct. The crack lengths must be greater than 0.
- ERROR: The number of axial stiffeners (NL) must be: $1 \leq NL \leq 9$
- ERROR: The number of crack lengths (Na) must be: $1 \leq Na \leq 50$
- ERROR: The fasteners position definition is not correct. The X and Y coordinates must be greater than 0.
- ERROR: The number of rivets (Ndr) must be: $1 \leq Ndr \leq 15$
- ERROR: The definition of the rivets position (Y coordinate) is not correct. The rivet position should be defined in such a way that their Y coordinate is arranged from the least to the greatest.

Error messages in the input data. Spanish output:

ERROR: No se puede abrir el fichero de entrada.
ERROR: Número de grupo de datos erróneo en el fichero de entrada.
ERROR: Demasiados rigidizadores, excediendo la capacidad del programa.
ERROR: Demasiadas longitudes de grieta para el análisis, excediendo la capacidad del programa.
ERROR: Demasiados remaches en un rigidizador, excediendo la capacidad del programa
ERROR: Grupo 1 ausente en el fichero de entrada (Título).
ERROR: Fin de fichero mientras se leía el grupo 1 en el fichero de entrada (Título).
ERROR: Grupo 2 ausente en el fichero de entrada (Panel).
ERROR: Fin de fichero mientras se leía el grupo 2 en el fichero de entrada (Panel).
ERROR: Grupo 2 mal definido (Revestimiento).
ERROR: Grupo 3 ausente en el fichero de entrada (Rigidizadores).
ERROR: Fin de fichero mientras se leía el grupo 3 en el fichero de entrada (Rigidizadores).
ERROR: Grupo 3 mal definido (Rigidizadores).
ERROR: Grupo 4 ausente en el fichero de entrada (Remaches).
ERROR: Fin de fichero mientras se leía el grupo 4 en el fichero de entrada (Remaches).
ERROR: Grupo 4 mal definido (Remaches).
ERROR: El grupo 4 (Remaches) no puede ser definido delante del grupo 3 (Rigidizadores).
ERROR: Tipo desconocido de distribución de remaches.
ERROR: Grupo 5 ausente en el fichero de entrada (Campo de esfuerzos).
ERROR: Fin de fichero mientras se leía el grupo 5 en el fichero de entrada (Campo de esfuerzos).
ERROR: Grupo 5 mal definido (Campo de esfuerzos).
ERROR: Grupo 6 ausente en el fichero de entrada (Longitudes de grieta).
ERROR: Fin de fichero mientras se leía el grupo 6 en el fichero de entrada (Longitudes de grieta).
ERROR: Grupo 6 mal definido (Longitudes de grieta).
ERROR: Grupo 7 ausente en el fichero de entrada (Control de impresión).
ERROR: Fin de fichero mientras se leía el grupo 7 en el fichero de entrada (Control de impresión).
ERROR: Grupo 7 mal definido (Control de impresión).
ERROR: No se puede abrir el fichero de salida.
ERROR: No se puede escribir en el fichero de salida.
ERROR: Las propiedades del material del panel no están bien definidas. Todas las propiedades deben tener un valor mayor de 0.
ERROR: La geometría del panel no están bien definida. El espesor del panel debe ser mayor de 0.
ERROR: Las propiedades del material de los rigidizadores del panel no están bien definidas. Todas las propiedades deben tener un valor mayor de 0.
ERROR: Los límites de rotura y fluencia del material del rigidizador no están bien definidos ($F_{tu} < F_{ty}$).
ERROR: La geometría de los rigidizadores no está bien definida. Todas las propiedades geométricas deben ser mayores de 0.
ERROR: La geometría de los rigidizadores no está bien definida ($A_{gross} < A_{net}$, $I_{gross} < I_{net}$).

ERROR: Las propiedades del material del remachado no están bien definidas.

ERROR: El paso entre remaches no está bien definido ($\text{paso} < \text{diámetro}$).

ERROR: La geometría del remachado no está bien definida. Todas las propiedades deben tener un valor mayor de 0.

ERROR: Las posiciones de los rigidizadores (coordenada X) deben definirse de menor a mayor.

ERROR: La distancia entre rigidizadores es menor al diámetro del remache.

ERROR: La posición de los remaches no está bien definida. Los remaches a ser considerados en el análisis no pueden estar más allá del primer apoyo transversal (L).

ERROR: La coordenada X de alguno de los remaches no es coherente con la coordenada X de los larguerillos.

ERROR: La geometría del panel no está bien definida. La distancia entre apoyo transversal (L) no puede ser menor de 0.

ERROR: El esfuerzo de referencia no está bien definido. El valor del esfuerzo de referencia debe ser mayor de 0.

ERROR: La posición de los rigidizadores no está bien definida. La coordenada X de los larguerillos debe ser mayor de 0.

ERROR: Las longitudes de grieta no están bien definidas. Las longitudes de grieta deben ser mayor de 0.

ERROR: El número de rigidizadores (NL) debe ser: $1 \leq NL \leq 9$.

ERROR: El número de longitudes de grieta (Na) debe ser: $1 \leq Na \leq 50$.

ERROR: La posición de los remaches no está bien definida. Las coordenadas X, Y de los remaches deben ser mayor de 0.

ERROR: El número de remaches (Ndr) debe ser: $1 \leq Ndr \leq 15$

ERROR: Las posiciones de los remaches (coordenada Y) debe definirse de menor a mayor.

5 LIMITATIONS AND ADDITIONAL CONSIDERATIONS

According to the theoretical manual of the REMACHADO program (Ref. 1), the following limitations and considerations have to be taken into account:

5.1 Symmetry

The skin panel – stiffeners assembly is supposed to be doubly symmetric and with a crack centred in the origin of the coordinate system.

Figure 9 and Figure 10 show the initial crack configuration of the typical problem to be solved with REMACHADO and the corresponding geometry to be analysed respectively.

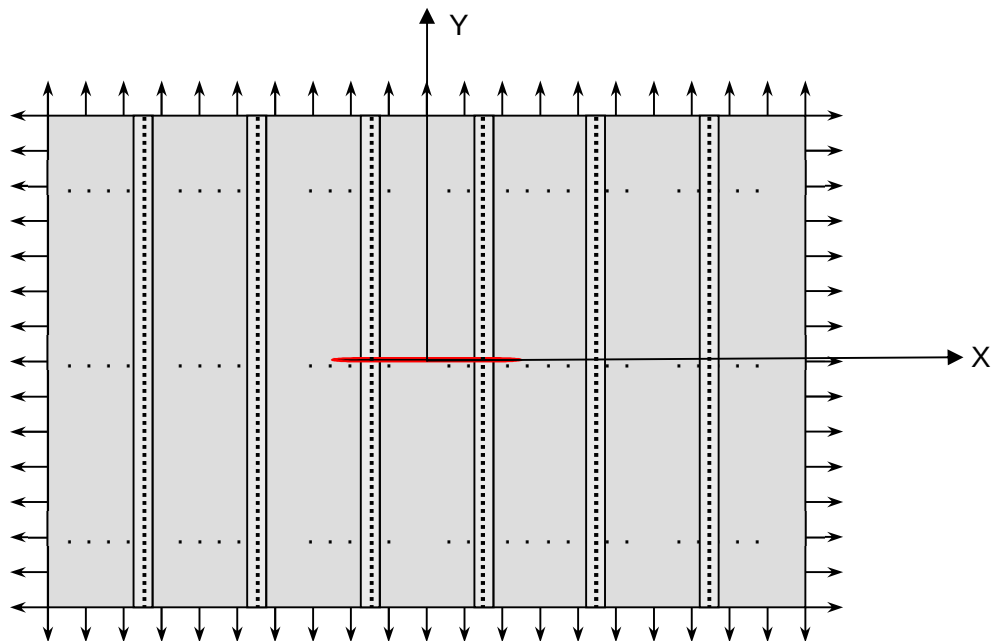


Figure 9. *Initial crack configuration*

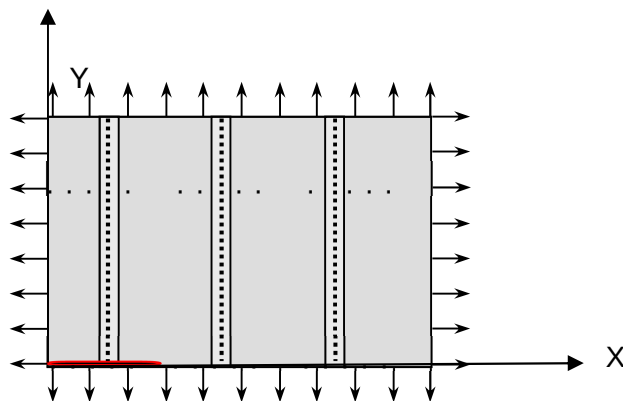


Figure 10. *Crack configuration analysed with REMACHADO*

For the non-symmetrical configurations analysis:

- No symmetry with respect to the crack propagation direction:

If the stringer geometry (area and inertia) is not constant between the transversal supports, the stringer section properties which stiffness is lower have to be used conservatively.

- No symmetry with respect to the normal to the crack propagation direction:

According to Ref. 1 and based on engineering judgment, two symmetrical problems have to be solved separately, one for each crack tip.

The normalized stress intensity factors (β factor) calculated for each crack tip have to be combined with the other factors calculated.

5.2 Longitudinal cracks

5.2.1 Distance between transversal supports

The bending stiffness of the stringers is calculated by the program REMACHADO by assuming that the stringers are supported on transversal supports such as frames or ribs, depending on the structural element considered (see Figure 1 and Figure 2). The bending stiffness of these transversal supports is infinite in comparison to the bending stiffness of the stringers.

For longitudinal cracks, where the crack grows parallel to the stringers, this hypothesis is not fulfilled because the transversal supports are the stringers and they have a very low bending stiffness compared with the frame (see Figure 11). According to Ref. 1, a huge (infinite) distance between stringers has to be considered in order to analyse this configuration properly. This is specified in the input file by using a null value of the distance between transversal support ($L = 0$).

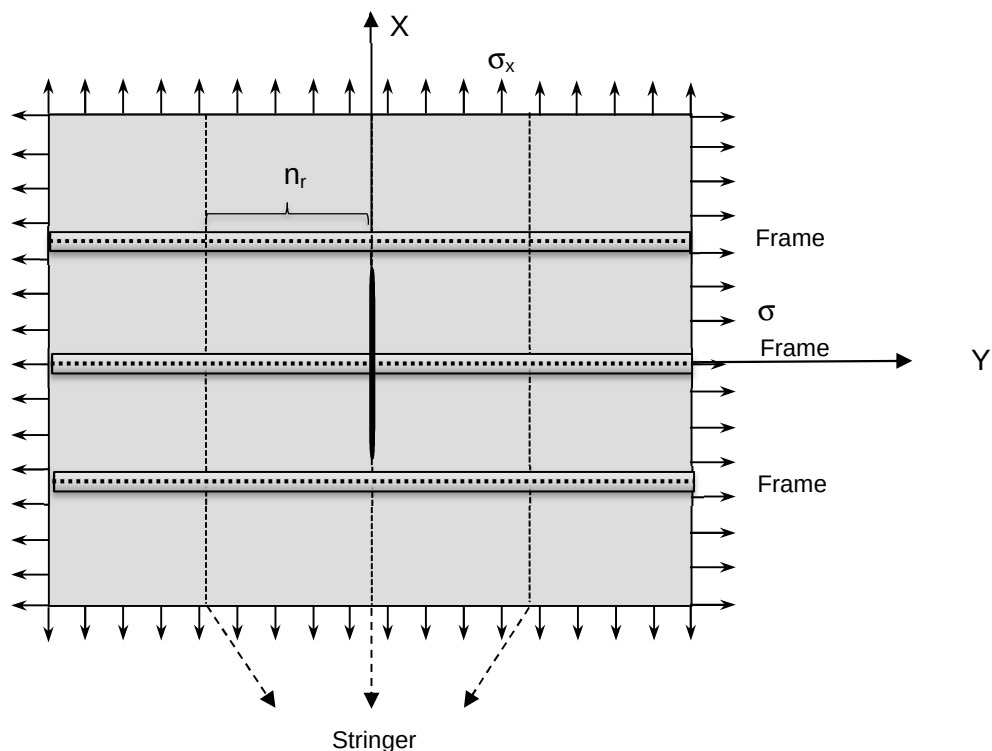


Figure 11. Initial crack configuration – Longitudinal crack in a fuselage skin panel

Note that the X axis is always coincident with the crack growing direction. Therefore, it has the same direction as the stringers in this analysis case.

5.2.2 Fasteners stiffness

For longitudinal cracks the fasteners defined in the Group 4 of the input file correspond with the joint of the frames to the skin (see Figure 11).

The fastener stiffness is either calculated by the program in accordance with the Huth expression (see Ref. 1 and section 3.2) or introduced manually by the user.

The joint considered by the program to calculate the stiffness is similar to the one shown in Figure 12:

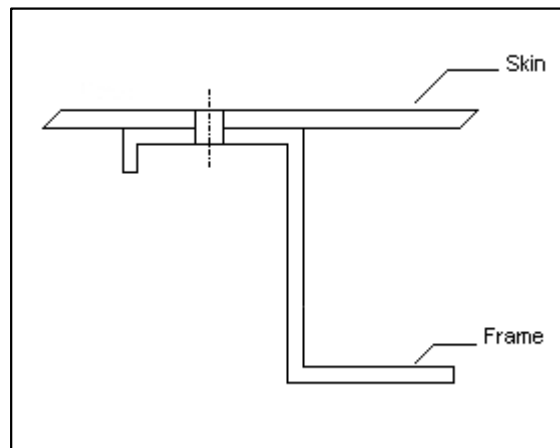


Figure 12. *Skin-frame joint*

In those cases where the joint between the skin and the frame is made with a clip (see Figure 13), the use of the Module 6 of the program UTILIDADES is highly recommended to calculate the rivet stiffness to be used for the analysis (see Ref. 3).

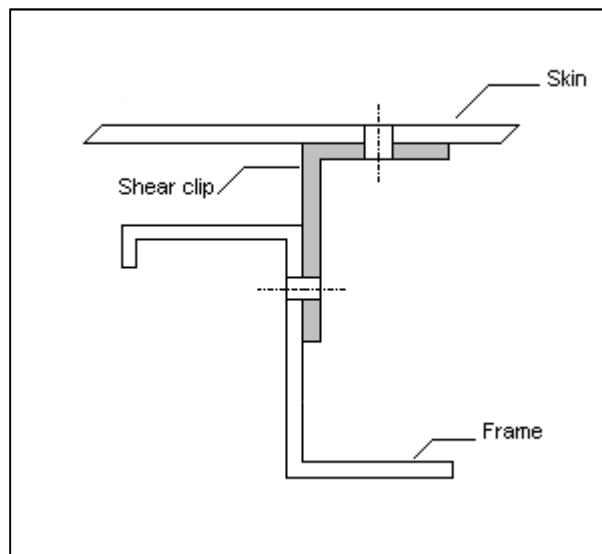


Figure 13. *Skin-frame joint with shear clip*

5.3 Transversal cracks

5.3.1 Distance between transversal supports

According to Ref. 1, the stringers' bending is calculated by assuming that the crack growing section coincides with the central transversal support (frame/rib) and one or two transversal supports are considered, one at each side of the crack, and equally spaced (see Figure 1 and Figure 2). The value of the distance between the transversal supports is defined in the input file through the parameter "L".

If the crack growing section does not coincide with the central stiffener (frame/rib), the hypotheses assumed for the stringers bending calculation are not fulfilled. In order to use the program REMACHADO properly to analyse this configuration, a huge (infinite) distance between transversal supports has to be considered. This is specified in the input file by using a value of $L = 0$ (see Ref. 1).

Figure 14 shows a transversal crack growing between two transversal supports where the horizontal red dotted lines are the transversal supports, and the vertical blue rectangles are the stringers.

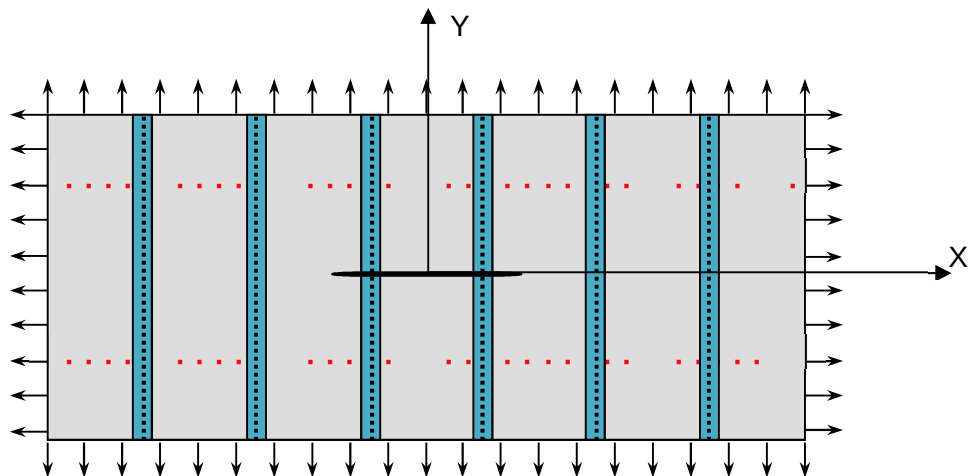


Figure 14. Crack between two transversal supports

I

5.4 Fasteners

The maximum number of fasteners considered by the program REMACHADO for the load transfer calculation is 15.

On the other hand, the fasteners have to be located between the crack section and the first transversal support. The program REMACHADO verifies this condition if $L \neq 0$. However, for those configurations where $L=0$ (see 5.2.1 and 5.3), the fastener position has to be checked by the user.

Note that a wrong fastener definition would lead to erroneous load transfer results and consequently to deviations in the stress intensity factor calculation.

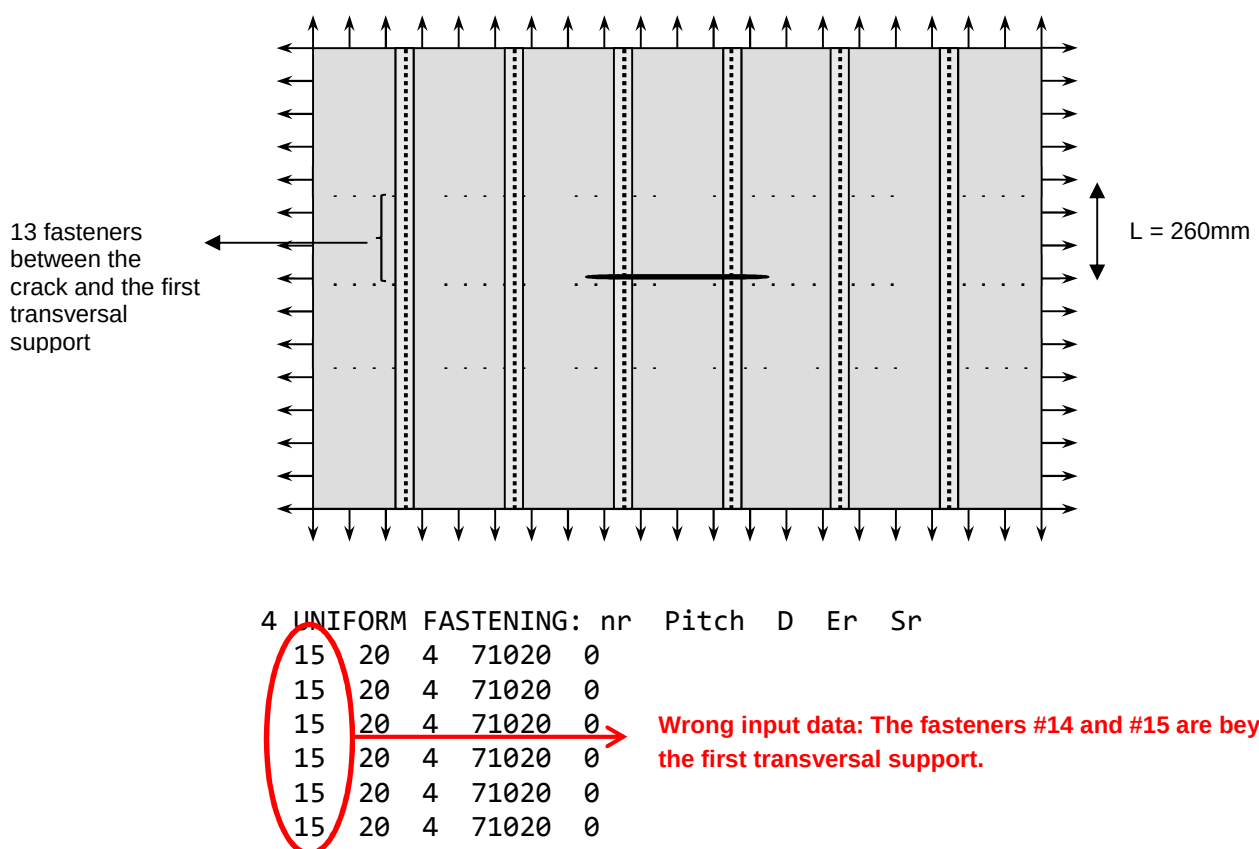


Figure 15. Wrong fastener position definition

6 EXAMPLES

Two examples are given below for the use of the program; both of them correspond to the same problem, presented in different forms.

6.1 Formulation of the example problem:

The problem consists of a 2024 T3 material panel, with a constant thickness, equal to 2 mm, with a crack in the origin of the coordinates.

The panel is stiffened by extruded stringers made of material 7075 T7653 and of a constant section, one of them is located at $X = 0$, and the rest are spaced at every 100 mm in a uniform manner. Figure 16 shows the stringer section:

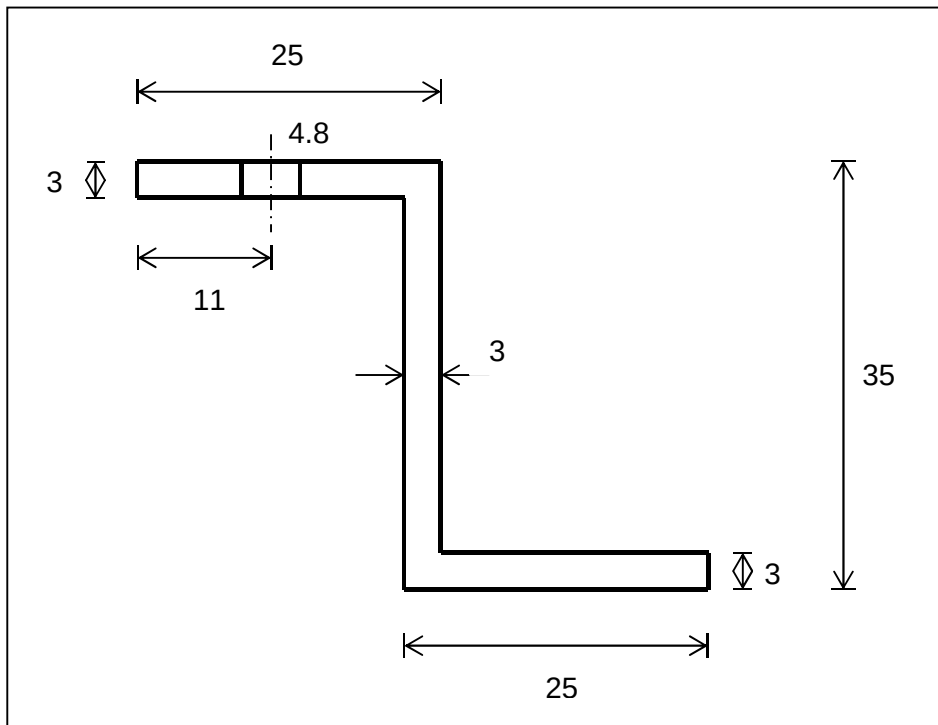


Figure 16. Example 1. Stringer geometry.

The stringers are attached to the skin by a row of fasteners of diameter 4.8 mm, uniformly spaced at every 20 mm.

The stringers, in turn, are resting on frames, which are supposed to be infinitely rigid. One of them coincides with the direction of propagation of the crack and they are all spaced at every 500 mm.

The panel – stringers set is submitted to a reference limit axial stress in the far field, of 200 N/mm². The stress in the normal direction to the stringers is half of the reference one.

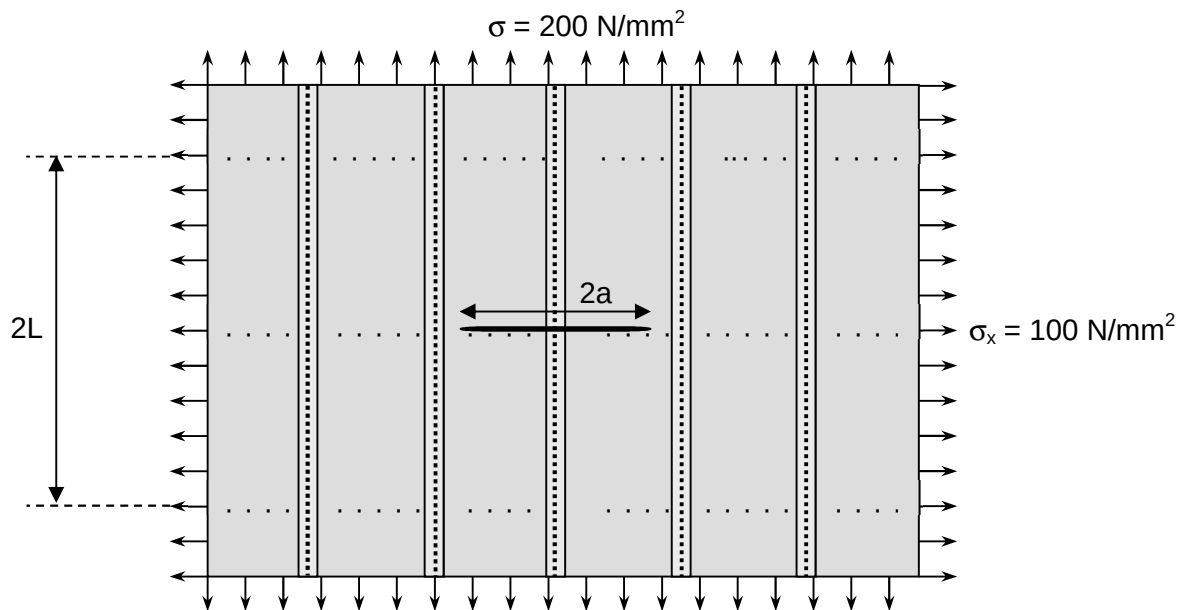


Figure 17. Example 1. General arrangement.

The data on the skin that must be introduced in the program are the following:

- Young modulus $E = 72400 \text{ N/mm}^2$
- Poisson modulus $\mu = 0.33$
- Thickness $t = 2 \text{ mm}$
- Fracture toughness $Kc = 5000 \text{ N/mm}^2 \cdot \sqrt{\text{mm}}$
- Spacing between transversal supports $L = 500 \text{ mm}$

The stringers are all of them equal, with the following properties:

- Thickness in the fastened flange $t_s = 3 \text{ mm}$
- Gross Area $A_g = 237 \text{ mm}^2$
- Net Area $A_n = 222.6 \text{ mm}^2$
- Gross moment of inertia $I_g = 44610 \text{ mm}^4$
- Net moment of inertia $I_n = 40674 \text{ mm}^4$
- Distance of CG to the exterior edge $Z_o = 17.5 \text{ mm}$
- Distance of the CG to the interior edge $Z_i = 17.5 \text{ mm}$
- Young's modulus $E_s = 71700 \text{ N/mm}^2$
- Ultimate tensile strength $F_{tu} = 469 \text{ N/mm}^2$
- Yield tensile stress $F_{ti} = 400 \text{ N/mm}^2$

The fasteners, installed uniformly for all the stringers, will have the following characteristics:

• Fasteners pitch	P	=	20	mm
• Diameter	D	=	4.8	mm
• Young's modulus	Er	=	71000	N/mm ²
• Stiffness	Sr	=	33865.65	N/mm

The far field of stress will be defined by the following:

• Reference stress	σ	=	200	N/mm ²
• Axial – lateral ratio	R	=	0.5	

The analysis will be carried out for half crack lengths between 0 and 200 mm. Therefore, it will only be necessary to consider the effect of the stringers located at: X = 0, X = 100 and X = 200 mm. Due to the symmetry, the ones located at X = -100 and X = -200 mm will implicitly be considered:

• Number de stiffeners	nl	=	3					
• Coordinates of the stiffeners	X	=	0	100	200	mm		
• Number of points of analysis	na	=	21					
• Half crack lengths	a	=	0	10	20	30	40	
	50	60	70	80	90	100	110	
	120	130	140	150	160	170	180	
	190	200						

6.2 Example 1

In this example, the previous problem will be solved with the following additional considerations, in the two runs of the program:

- Run 1:

Fifteen fasteners (at each side of the crack) per stiffener will be considered for the load transfer between the stiffeners and the skin.

In the input data, a uniform fastening will be specified with the data previously defined.

The program, by means of the expression given in chapter 3.2, will automatically calculate the stiffness of the fasteners. For this purpose, a null stiffness will be specified in the input data.

All the stiffeners will be considered initially intact and the program will be allowed to determine the failure.

The results requested will be all the possible ones, that is, the results for the skin, for the stiffeners, allowable stresses and fastener loads.

- Run 2:

Five fasteners (at each side of the crack) per stiffener will be considered for the load transfer between the stiffeners and the skin.

In the input data, a variable fastening will be specified with the data defined previously, and so, in reality, it will be in the same way as in the first step.

The stiffness of the fasteners will be specified in the data input, as previously defined, which coincides with the one calculated by the program in the first step.

All the stiffeners will be considered initially intact and the program will be allowed to determine the failure.

Results will only be requested for the skin.

The difference between both runs lies in the definition of the fastening, although it will formally be the same in both cases, except for the number of fasteners.

Therefore, this example could serve to evaluate the influence of the total number of fasteners considered per stringer.

With this data, we can prepare the input file. This file as well as the output file obtained by running the program is shown on next pages:

Input file:

```

1 Program REMACHADO. Example 1.
  Crack with central stiffener intact.
  Material: Skin: 2024 T3. Stiffeners: 7075 T73511.
  Uniform fastening. 15 fasteners per stringer.
2 SKIN DATA: E  μ  t  Kc  L
  72400.0      0.33  2.0      5000.0      500.0
3 STRINGERS DATA: nl  X  ts  Ag  An  Ig  In  Zo  Zi  Es  FtU  Fty  Fail
  3      0.0      100.0      200.0
      3.0      3.0      3.0
      237.0      237.0      237.0
      222.6      222.6      222.6
      44610.0      44610.0      44610.0
      40674.0      40674.0      40674.0
      17.5      17.5      17.5
      17.5      17.5      17.5
      71700.0      71700.0      71700.0
      469.0      469.0      469.0
      400.0      400.0      400.0
      0      0      0
4 UNIFORM FASTENING: nr  Pitch  D  Er  Sr
  15      20.00      4.80      71000.00      0.00
  15      20.00      4.80      71000.00      0.00
  15      20.00      4.80      71000.00      0.00
5 FAR FIELD STRESS: Reference_stress  Sx/Sy
  200.00      0.50
6 CRACK LENGTHS
  21  0.0  10.0  20.0  30.0  40.0  50.0
      60.0  70.0  80.0  90.0  100.0
      110.0  120.0  130.0  140.0  150.0
      160.0  170.0  180.0  190.0  200.0
7 PRINT CONTROL: Skin  Stringers  Allowables  Forces
  1  1  1  1
8 RUN THE PROGRAM
1 Program REMACHADO. Example 1.
  Crack with central stiffener intact.
  Material: Skin: 2024 T3. Stiffeners: 7075 T73511.
  Variable fastening. 5 fasteners per stringer.
4 VARIABLE FASTENING: n  X(i)  Y(i)  D(i)  E(i)  K(i)
  5      0.00      10.00      4.80      71000.00      33865.65
      0.00      30.00      4.80      71000.00      33865.65
      0.00      50.00      4.80      71000.00      33865.65
      0.00      70.00      4.80      71000.00      33865.65
      0.00      90.00      4.80      71000.00      33865.65
  5      100.00      10.00      4.80      71000.00      33865.65
      100.00      30.00      4.80      71000.00      33865.65
      100.00      50.00      4.80      71000.00      33865.65
      100.00      70.00      4.80      71000.00      33865.65
      100.00      90.00      4.80      71000.00      33865.65
  5      200.00      10.00      4.80      71000.00      33865.65
      200.00      30.00      4.80      71000.00      33865.65
      200.00      50.00      4.80      71000.00      33865.65
      200.00      70.00      4.80      71000.00      33865.65
      200.00      90.00      4.80      71000.00      33865.65
7 PRINT CONTROL: Skin  Stringers  Allowables  Forces
  1  0  0  0
8 RUN THE PROGRAM

```

Output file:

PAGE 1

Program REMACHADO. Example 1.
Crack with central stiffener intact.
Material: Skin: 2024 T3. Stiffeners: 7075 T73511.
Uniform fastening. 15 fasteners per stringer.

```
*****
*                                     *
*   <<< I N P U T   D A T A >>>   *
*                                     *
*****
```

EXECUTION DATA:

=====

Date: 16-Mar-2018
Remachado 3.2. (c) Airbus Defence and Space. Mar-2018

SKIN DATA:

=====

Young's modulus.....:	72400.00
Poisson's ratio.....:	0.33
Thickness.....:	2.00
Fracture toughness Kc.....:	5000.00
Transversal support spacing...:	500.00

APPLIED STRESSES:

=====

Average axial stress.....:	200.00
Lateral-axial stress ratio...:	0.50

STIFFENERS DATA:

=====

	STIFFENER 1	STIFFENER 2	STIFFENER 3
	-----	-----	-----
X-coordinate.....:	0.00	100.00	200.00
Uniform fastener system:			
Number of fasteners.....:	15	15	15
Fastener pitch.....:	20.00	20.00	20.00
Fastener Young's modulus..:	71000.00	71000.00	71000.00
Fastener stiffness.....:	0.00	0.00	0.00
Fastener diameter.....:	4.80	4.80	4.80
Thickness at attach point.:	3.00	3.00	3.00
Gross area.....:	237.00	237.00	237.00
Net area.....:	222.60	222.60	222.60
Gross moment of inertia.....:	44610.00	44610.00	44610.00
Net moment of inertia.....:	40674.00	40674.00	40674.00
Distance from CG to outer cap:	17.50	17.50	17.50
Distance from CG to inner cap:	17.50	17.50	17.50
Young's modulus.....:	71700.00	71700.00	71700.00
Ftu of stiffener.....:	469.00	469.00	469.00
Fty of stiffener.....:	400.00	400.00	400.00
Initial state of stiffener....:	Intact	Intact	Intact

Output file (cont.):

Program REMACHADO. Example 1.
Crack with central stiffener intact.
Material: Skin: 2024 T3. Stiffeners: 7075 T73511.
Uniform fastening. 15 fasteners per stringer.

PAGE 2

```
*****
*                                     *
*   <<< R E S U L T S   F O R   S K I N >>>   *
*                                     *
*****
```

APPLIED STRESSES:

=====

Average axial stress (YD).....: 200.00
Ratio of stresses (XD/YD).....: 0.50
Unit axial skin stress.....: 218.87
Unit lateral skin stress.....: 100.00

HALF CRACK LENGTH	STRESS IN THE FAR FIELD	BETA FACTOR	STRESS INTENSITY FACTOR	STIFFENERS STATUS
-----	-----	-----	-----	-----
0.00	218.87	1.00000	0.00	III
10.00	218.87	0.92493	1134.68	III
20.00	218.87	0.88123	1528.86	III
30.00	218.87	0.85555	1817.89	III
40.00	218.87	0.83686	2053.28	III
50.00	218.87	0.82160	2253.78	III
60.00	218.87	0.80748	2426.44	III
70.00	218.87	0.79244	2572.06	III
80.00	218.87	0.77364	2684.41	III
90.00	218.87	0.74460	2740.36	III
100.00	218.87	0.66440	2577.46	III
110.00	218.87	0.57779	2350.88	III
120.00	218.87	0.56228	2389.50	III
130.00	218.87	1.57790	6979.36	XXI
140.00	218.87	1.51656	6961.27	XXI
150.00	218.87	1.46231	6947.83	XXI
160.00	218.87	1.41043	6921.10	XXI
170.00	218.87	1.35693	6863.52	XXI
180.00	218.87	1.29654	6748.20	XXI
190.00	218.87	1.21795	6512.85	XXI
200.00	218.87	1.98961	10915.62	XXX

Beta factors referred to the unit axial skin stress.

Key for stiffeners status: I=Intact
X=Failed

Output file (cont.):

Program REMACHADO. Example 1.
Crack with central stiffener intact.
Material: Skin: 2024 T3. Stiffeners: 7075 T73511.
Uniform fastening. 15 fasteners per stringer.

PAGE 3

```
*****
*
*   <<< R E S U L T S   F O R   S T I F F E N E R   A T   X =   0.00 >>>
*
*****
```

Unit stiffener stress: 184.07

HALF CRACK LENGTH	AXIAL FORCE	BENDING MOMENT	GROSS STRESS		NET STRESS	
			OUTER CAP	INNER CAP	OUTER CAP	INNER CAP
0.00	43625.7	0.0	184.07	184.07	195.98	195.98
10.00	44950.5	20641.2	197.76	181.57	210.81	193.05
20.00	47319.1	55776.3	221.54	177.78	236.57	188.58
30.00	49977.4	93164.9	247.42	174.33	264.60	184.43
40.00	52750.1	130330.4	273.70	171.45	293.05	180.90
50.00	55551.2	166242.7	299.61	169.18	321.08	178.03
60.00	58327.8	200386.2	324.72	167.50	348.25	175.81
70.00	61035.9	232401.8	348.70	166.37	374.19	174.20
80.00	63629.1	261936.6	371.23	165.72	398.54	173.15
90.00	66041.9	288455.4	391.82	165.50	420.79	172.58
100.00	68110.0	310414.0	409.16	165.61	439.53	172.42
110.00	69557.8	325193.9	421.06	165.92	452.39	172.56
120.00	70737.1	336688.3	430.55	166.39	462.64	172.92
130.00	0.0	0.0	0.00	0.00	0.00	0.00
140.00	0.0	0.0	0.00	0.00	0.00	0.00
150.00	0.0	0.0	0.00	0.00	0.00	0.00
160.00	0.0	0.0	0.00	0.00	0.00	0.00
170.00	0.0	0.0	0.00	0.00	0.00	0.00
180.00	0.0	0.0	0.00	0.00	0.00	0.00
190.00	0.0	0.0	0.00	0.00	0.00	0.00
200.00	0.0	0.0	0.00	0.00	0.00	0.00

Loads and stresses for stiffener at Y = 0.

Output file (cont.):

Program REMACHADO. Example 1. PAGE 4
Crack with central stiffener intact.
Material: Skin: 2024 T3. Stiffeners: 7075 T73511.
Uniform fastening. 15 fasteners per stringer.

*
* <<< R E S U L T S F O R S T I F F E N E R A T X = 100.00 >>> *
*

Unit stiffener stress: 184.07

HALF CRACK LENGTH	AXIAL FORCE	BENDING MOMENT	GROSS STRESS		NET STRESS	
			OUTER CAP	INNER CAP	OUTER CAP	INNER CAP
0.00	43625.7	0.0	184.07	184.07	195.98	195.98
10.00	43696.3	339.2	184.51	184.24	196.45	196.15
20.00	43890.5	1292.7	185.70	184.69	197.73	196.62
30.00	44206.8	2938.9	187.68	185.37	199.86	197.33
40.00	44656.9	5497.4	190.58	186.27	202.98	198.25
50.00	45263.7	9345.7	194.65	187.32	207.36	199.32
60.00	46064.1	15092.8	200.28	188.44	213.43	200.44
70.00	47116.3	23731.5	208.11	189.49	221.87	201.45
80.00	48518.1	36996.2	219.23	190.20	233.88	202.04
90.00	50475.9	58659.7	235.99	189.97	251.99	201.52
100.00	53589.1	100040.4	265.36	186.87	283.78	197.70
110.00	57174.2	151642.0	300.73	181.75	322.09	191.60
120.00	60040.3	191430.6	328.43	178.24	352.09	187.36
130.00	0.0	0.0	0.00	0.00	0.00	0.00
140.00	0.0	0.0	0.00	0.00	0.00	0.00
150.00	0.0	0.0	0.00	0.00	0.00	0.00
160.00	0.0	0.0	0.00	0.00	0.00	0.00
170.00	0.0	0.0	0.00	0.00	0.00	0.00
180.00	0.0	0.0	0.00	0.00	0.00	0.00
190.00	0.0	0.0	0.00	0.00	0.00	0.00
200.00	0.0	0.0	0.00	0.00	0.00	0.00

Loads and stresses for stiffener at Y = 0.

Output file (cont.):

Program REMACHADO. Example 1.
Crack with central stiffener intact.
Material: Skin: 2024 T3. Stiffeners: 7075 T73511.
Uniform fastening. 15 fasteners per stringer.

PAGE 5

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*****  
*  
*   <<< R E S U L T S   F O R   S T I F F E N E R   A T   X = 200.00 >>>   *  
*  
*****
```

Unit stiffener stress: 184.07

HALF CRACK LENGTH	AXIAL FORCE	BENDING MOMENT	GROSS STRESS		NET STRESS	
			OUTER CAP	INNER CAP	OUTER CAP	INNER CAP
0.00	43625.7	0.0	184.07	184.07	195.98	195.98
10.00	43646.4	5.4	184.16	184.16	196.08	196.07
20.00	43702.2	14.2	184.40	184.39	196.33	196.32
30.00	43790.5	25.8	184.78	184.76	196.73	196.71
40.00	43911.1	48.4	185.30	185.26	197.29	197.24
50.00	44065.3	98.4	185.97	185.89	198.00	197.91
60.00	44254.9	201.3	186.81	186.65	198.90	198.72
70.00	44482.1	393.0	187.84	187.53	200.00	199.66
80.00	44748.5	721.0	189.10	188.53	201.34	200.72
90.00	45053.1	1243.1	190.59	189.61	202.93	201.86
100.00	45377.2	1992.6	192.25	190.68	204.71	202.99
110.00	45669.7	2887.8	193.83	191.57	206.41	203.92
120.00	45985.8	4111.2	195.65	192.42	208.35	204.82
130.00	61388.7	31891.6	271.53	246.51	289.50	262.06
140.00	63894.4	41743.1	285.97	253.22	305.00	269.08
150.00	66696.2	55051.5	303.01	259.82	323.31	275.94
160.00	69887.6	73287.6	323.63	266.13	345.49	282.43
170.00	73597.0	98709.2	349.26	271.81	373.09	288.15
180.00	78025.8	135100.1	382.22	276.22	408.65	292.39
190.00	83592.3	190558.2	427.46	277.96	457.51	293.54
200.00	0.0	0.0	0.00	0.00	0.00	0.00

Loads and stresses for stiffener at Y = 0.

Output file (cont.):

Program REMACHADO. Example 1.
Crack with central stiffener intact.
Material: Skin: 2024 T3. Stiffeners: 7075 T73511.
Uniform fastening. 15 fasteners per stringer.

PAGE 6

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*****
*                                     *
*   <<< A L L O W E D   S T R E S S E S >>>   *
*                                     *
*****
```

HALF CRACK LENGTH	SKIN	STIFFENER 1		STIFFENER 2		STIFFENER 3	
		Smax=Ftu	Smed=Fty	Smax=Ftu	Smed=Fty	Smax=Ftu	Smed=Fty
0.00	*****	478.61	408.20	478.61	408.20	478.61	408.20
10.00	881.30	444.94	396.17	477.49	407.54	478.38	408.01
20.00	654.08	396.50	376.34	474.39	405.74	477.76	407.48
30.00	550.09	354.50	356.32	469.33	402.83	476.79	406.66
40.00	487.03	320.09	337.59	462.11	398.77	475.45	405.55
50.00	443.70	292.14	320.57	452.35	393.43	473.74	404.13
60.00	412.13	269.35	305.31	439.49	386.59	471.60	402.40
70.00	388.79	250.68	291.76	422.76	377.96	469.00	400.34
80.00	372.52	235.36	279.87	401.06	367.04	465.89	397.96
90.00	364.92	222.91	269.65	372.23	352.80	462.23	395.27
100.00	387.98	213.41	261.46	330.53	332.31	458.21	392.44
110.00	425.37	207.34	256.02	291.22	311.47	454.44	389.93
120.00	418.50	202.75	251.75	266.41	296.60	450.20	387.25
130.00	143.28	0.00	0.00	0.00	0.00	324.00	290.09
140.00	143.65	0.00	0.00	0.00	0.00	307.54	278.71
150.00	143.93	0.00	0.00	0.00	0.00	290.12	267.00
160.00	144.49	0.00	0.00	0.00	0.00	271.50	254.81
170.00	145.70	0.00	0.00	0.00	0.00	251.41	241.97
180.00	148.19	0.00	0.00	0.00	0.00	229.54	228.23
190.00	153.54	0.00	0.00	0.00	0.00	205.02	213.03
200.00	91.61	0.00	0.00	0.00	0.00	0.00	0.00

Allowed stresses are based on the reference stress.

Output file (cont.):

Program REMACHADO. Example 1.
Crack with central stiffener intact.
Material: Skin: 2024 T3. Stiffeners: 7075 T73511.
Uniform fastening. 15 fasteners per stringer.

PAGE 7

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*****
*
*   <<< F O R C E S   O N   F A S T E N E R S >>>
*
*****
```

Half crack length: 0.000

FASTENER	X = 0.00	X = 100.00	X = 200.00
-----	-----	-----	-----
1	-0.0	-0.0	-0.0
2	0.0	0.0	0.0
3	0.0	0.0	0.0
4	-0.0	-0.0	-0.0
5	0.0	-0.0	0.0
6	0.0	0.0	0.0
7	0.0	0.0	0.0
8	0.0	0.0	-0.0
9	0.0	0.0	-0.0
10	-0.0	-0.0	-0.0
11	-0.0	-0.0	0.0
12	-0.0	-0.0	-0.0
13	0.0	0.0	0.0
14	0.0	0.0	-0.0
15	0.0	-0.0	0.0

Positive fastener forces induce compression in the skin.

Output file (cont.):

Program REMACHADO. Example 1.
Crack with central stiffener intact.
Material: Skin: 2024 T3. Stiffeners: 7075 T73511.
Uniform fastening. 15 fasteners per stringer.

PAGE 8

```
*****
*
*   <<< F O R C E S   O N   F A S T E N E R S >>>
*
*****
```

Half crack length: 10.000

FASTENER	X = 0.00	X = 100.00	X = 200.00
-----	-----	-----	-----
1	891.5	-0.1	-0.1
2	268.6	0.8	-0.1
3	106.6	3.5	-0.1
4	45.9	6.3	0.1
5	19.3	8.1	0.5
6	6.7	8.6	0.9
7	0.8	8.1	1.2
8	-1.9	7.1	1.6
9	-2.9	6.1	1.8
10	-3.0	5.1	2.0
11	-2.7	4.2	2.1
12	-2.1	3.6	2.3
13	-1.4	3.2	2.5
14	-0.7	3.0	2.8
15	0.2	3.1	3.3

Positive fastener forces induce compression in the skin.

Output file (cont.):

Program REMACHADO. Example 1.
Crack with central stiffener intact.
Material: Skin: 2024 T3. Stiffeners: 7075 T73511.
Uniform fastening. 15 fasteners per stringer.

PAGE 9

```
*****
*
*   <<< F O R C E S   O N   F A S T E N E R S >>>
*
*****
```

Half crack length: 20.000

FASTENER	X = 0.00	X = 100.00	X = 200.00
-----	-----	-----	-----
1	2081.0	-0.6	-0.3
2	903.2	3.0	-0.6
3	407.2	13.5	-0.5
4	192.0	24.6	0.3
5	89.9	31.1	1.6
6	39.0	32.5	3.1
7	13.2	30.4	4.6
8	0.4	26.6	5.9
9	-5.5	22.5	6.8
10	-7.5	18.7	7.5
11	-7.4	15.6	8.0
12	-6.2	13.2	8.5
13	-4.3	11.6	9.2
14	-2.1	10.9	10.2
15	0.6	11.3	12.2

Positive fastener forces induce compression in the skin.

Output file (cont.):

Program REMACHADO. Example 1.
Crack with central stiffener intact.
Material: Skin: 2024 T3. Stiffeners: 7075 T73511.
Uniform fastening. 15 fasteners per stringer.

PAGE 10

```
*****
*
*   <<< F O R C E S   O N   F A S T E N E R S >>>
*
*****
```

Half crack length: 30.000

FASTENER	X = 0.00	X = 100.00	X = 200.00
-----	-----	-----	-----
1	3088.9	-1.3	-0.6
2	1649.7	7.9	-1.6
3	839.8	32.8	-1.3
4	429.5	57.1	0.5
5	217.9	70.1	3.5
6	105.9	71.7	6.9
7	45.9	66.1	10.1
8	14.2	57.4	12.8
9	-1.7	48.1	14.8
10	-8.8	39.7	16.3
11	-10.7	32.9	17.4
12	-9.8	27.7	18.4
13	-7.1	24.4	19.7
14	-3.3	22.8	22.0
15	1.3	23.7	26.1

Positive fastener forces induce compression in the skin.

Output file (cont.):

Program REMACHADO. Example 1.
Crack with central stiffener intact.
Material: Skin: 2024 T3. Stiffeners: 7075 T73511.
Uniform fastening. 15 fasteners per stringer.

PAGE 11

```
*****
*
*   <<< F O R C E S   O N   F A S T E N E R S >>>
*
*****
```

Half crack length: 40.000

FASTENER	X = 0.00	X = 100.00	X = 200.00
-----	-----	-----	-----
1	3956.6	-1.9	-1.2
2	2372.4	18.8	-3.0
3	1332.5	67.2	-2.5
4	732.8	109.0	0.8
5	397.0	128.0	6.2
6	208.5	127.2	12.2
7	102.6	115.0	17.9
8	43.6	98.5	22.5
9	11.9	81.8	26.0
10	-3.9	67.1	28.3
11	-10.3	55.3	30.1
12	-11.0	46.5	31.8
13	-8.3	40.8	34.0
14	-3.4	38.3	37.7
15	3.3	39.7	44.7

Positive fastener forces induce compression in the skin.

Output file (cont.):

Program REMACHADO. Example 1.
Crack with central stiffener intact.
Material: Skin: 2024 T3. Stiffeners: 7075 T73511.
Uniform fastening. 15 fasteners per stringer.

PAGE 12

```
*****
*
*   <<< F O R C E S   O N   F A S T E N E R S >>>
*
*****
```

Half crack length: 50.000

FASTENER	X = 0.00	X = 100.00	X = 200.00
-----	-----	-----	-----
1	4723.4	-1.2	-2.0
2	3031.1	43.5	-4.9
3	1833.3	127.6	-3.9
4	1073.1	188.2	1.5
5	614.8	208.5	10.1
6	343.1	199.7	19.6
7	183.2	176.3	28.4
8	90.1	148.6	35.4
9	37.4	122.1	40.4
10	9.1	99.5	43.8
11	-4.2	81.5	46.2
12	-8.3	68.4	48.5
13	-6.6	60.0	51.7
14	-1.2	56.4	57.2
15	7.2	58.8	67.6

Positive fastener forces induce compression in the skin.

Output file (cont.):

Program REMACHADO. Example 1.
Crack with central stiffener intact.
Material: Skin: 2024 T3. Stiffeners: 7075 T73511.
Uniform fastening. 15 fasteners per stringer.

PAGE 13

```
*****
*
*   <<< F O R C E S   O N   F A S T E N E R S >>>
*
*****
```

Half crack length: 60.000

FASTENER	X = 0.00	X = 100.00	X = 200.00
-----	-----	-----	-----
1	5412.5	3.7	-3.1
2	3622.0	98.0	-7.2
3	2313.2	231.0	-5.4
4	1425.8	304.1	3.1
5	857.0	314.9	15.8
6	502.4	289.3	29.7
7	284.7	248.7	42.1
8	153.0	205.9	51.7
9	75.2	167.2	58.3
10	31.3	135.2	62.6
11	8.6	110.3	65.6
12	-0.6	92.4	68.5
13	-1.2	81.1	72.7
14	4.0	76.6	80.2
15	14.0	80.3	94.5

Positive fastener forces induce compression in the skin.

Output file (cont.):

Program REMACHADO. Example 1.
Crack with central stiffener intact.
Material: Skin: 2024 T3. Stiffeners: 7075 T73511.
Uniform fastening. 15 fasteners per stringer.

PAGE 14

```
*****
*
*   <<< F O R C E S   O N   F A S T E N E R S >>>
*
*****
```

Half crack length: 70.000

FASTENER	X = 0.00	X = 100.00	X = 200.00
-----	-----	-----	-----
1	6035.5	21.3	-4.3
2	4150.0	216.2	-9.8
3	2757.9	401.4	-6.6
4	1772.4	465.6	6.0
5	1109.5	448.2	24.2
6	677.8	393.7	43.2
7	402.6	329.0	59.5
8	230.2	267.4	71.8
9	124.7	214.5	79.9
10	62.5	172.0	84.9
11	28.5	139.8	88.2
12	12.6	117.1	91.5
13	8.7	103.1	96.7
14	13.1	97.9	106.2
15	24.3	103.4	124.9

Positive fastener forces induce compression in the skin.

Output file (cont.):

Program REMACHADO. Example 1.
Crack with central stiffener intact.
Material: Skin: 2024 T3. Stiffeners: 7075 T73511.
Uniform fastening. 15 fasteners per stringer.

PAGE 15

```
*****
*
*   <<< F O R C E S   O N   F A S T E N E R S >>>
*
*****
```

Half crack length: 80.000

FASTENER	X = 0.00	X = 100.00	X = 200.00
-----	-----	-----	-----
1	6596.8	82.5	-5.7
2	4619.0	466.4	-12.6
3	3160.0	665.2	-6.8
4	2099.5	676.3	11.4
5	1359.5	604.6	36.1
6	859.8	507.8	60.9
7	530.4	412.3	81.4
8	317.7	328.8	96.0
9	183.6	260.5	105.2
10	101.9	207.4	110.4
11	55.1	168.1	113.7
12	31.5	140.9	117.1
13	23.4	124.8	123.1
14	26.4	119.5	134.7
15	38.8	127.6	158.0

Positive fastener forces induce compression in the skin.

Output file (cont.):

Program REMACHADO. Example 1.
Crack with central stiffener intact.
Material: Skin: 2024 T3. Stiffeners: 7075 T73511.
Uniform fastening. 15 fasteners per stringer.

PAGE 16

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*****
*
*   <<< F O R C E S   O N   F A S T E N E R S >>>
*
*****
```

Half crack length: 90.000

FASTENER	X = 0.00	X = 100.00	X = 200.00
-----	-----	-----	-----
1	7094.3	339.2	-7.2
2	5028.2	953.3	-14.8
3	3513.2	1032.2	-5.2
4	2395.4	925.3	20.3
5	1594.5	771.9	52.8
6	1037.7	621.4	83.7
7	660.3	490.9	107.9
8	410.2	384.5	124.2
9	248.5	301.2	133.7
10	147.3	238.3	138.5
11	87.4	192.8	141.2
12	55.4	162.2	144.3
13	42.6	144.9	150.9
14	43.9	140.4	164.6
15	57.3	151.9	192.6

Positive fastener forces induce compression in the skin.

Output file (cont.):

Program REMACHADO. Example 1.
Crack with central stiffener intact.
Material: Skin: 2024 T3. Stiffeners: 7075 T73511.
Uniform fastening. 15 fasteners per stringer.

PAGE 17

```
*****
*
*   <<< F O R C E S   O N   F A S T E N E R S >>>
*
*****
```

Half crack length: 100.000

FASTENER	X = 0.00	X = 100.00	X = 200.00
-----	-----	-----	-----
1	7504.1	1579.0	-8.4
2	5359.8	1682.8	-15.8
3	3799.8	1444.0	-0.5
4	2640.0	1168.6	33.7
5	1794.5	920.8	74.3
6	1194.1	715.4	110.6
7	778.4	552.1	137.5
8	497.3	425.8	154.4
9	312.0	330.4	163.2
10	193.6	260.3	166.8
11	121.6	210.8	168.4
12	81.8	178.6	170.9
13	64.5	161.4	177.7
14	64.1	158.9	193.1
15	78.6	174.6	225.6

Positive fastener forces induce compression in the skin.

Output file (cont.):

Program REMACHADO. Example 1.
Crack with central stiffener intact.
Material: Skin: 2024 T3. Stiffeners: 7075 T73511.
Uniform fastening. 15 fasteners per stringer.

PAGE 18

```
*****
*
*   <<< F O R C E S   O N   F A S T E N E R S >>>
*
*****
```

Half crack length: 110.000

FASTENER	X = 0.00	X = 100.00	X = 200.00
-----	-----	-----	-----
1	7778.9	3567.2	-9.2
2	5579.8	2414.1	-15.0
3	3989.0	1798.5	7.4
4	2803.4	1358.7	50.6
5	1931.4	1028.0	98.0
6	1304.7	777.9	137.9
7	865.0	589.7	165.5
8	563.6	449.4	181.5
9	362.1	346.2	188.6
10	231.7	272.1	190.5
11	151.0	220.9	190.5
12	105.4	188.5	192.1
13	84.7	172.5	198.8
14	82.9	172.4	215.5
15	98.5	192.4	251.3

Positive fastener forces induce compression in the skin.

Output file (cont.):

Program REMACHADO. Example 1.
Crack with central stiffener intact.
Material: Skin: 2024 T3. Stiffeners: 7075 T73511.
Uniform fastening. 15 fasteners per stringer.

PAGE 19

```
*****
*
*   <<< F O R C E S   O N   F A S T E N E R S >>>
*
*****
```

Half crack length: 120.000

FASTENER	X = 0.00	X = 100.00	X = 200.00
-----	-----	-----	-----
1	7991.8	4832.9	-10.0
2	5748.6	3127.1	-12.5
3	4133.0	2167.9	20.5
4	2929.2	1558.8	74.8
5	2039.7	1140.1	129.0
6	1395.2	842.2	170.8
7	938.4	627.5	197.2
8	622.0	472.3	210.5
9	408.2	361.0	214.6
10	267.9	282.6	213.6
11	180.1	229.7	211.5
12	129.4	197.2	211.7
13	105.6	182.3	218.1
14	102.8	184.5	235.8
15	119.6	208.5	274.5

Positive fastener forces induce compression in the skin.

Output file (cont.):

Program REMACHADO. Example 1.
Crack with central stiffener intact.
Material: Skin: 2024 T3. Stiffeners: 7075 T73511.
Uniform fastening. 15 fasteners per stringer.

PAGE 20

```
*****
*
*   <<< F O R C E S   O N   F A S T E N E R S >>>
*
*****
```

Half crack length: 130.000

FASTENER	X = 0.00	X = 100.00	X = 200.00
-----	-----	-----	-----
1	-1876.8	-3616.6	1.1
2	-1652.3	-3671.7	85.1
3	-1673.4	-3388.6	324.9
4	-1820.0	-2960.9	636.3
5	-2008.7	-2582.2	923.3
6	-2192.0	-2306.6	1142.3
7	-2354.4	-2135.1	1290.1
8	-2502.7	-2056.5	1381.3
9	-2657.4	-2061.6	1435.6
10	-2847.3	-2149.0	1473.5
11	-3109.6	-2327.8	1515.3
12	-3493.0	-2620.9	1583.2
13	-4068.5	-3073.0	1706.2
14	-4956.1	-3769.9	1928.1
15	-6413.7	-4905.4	2336.7

Positive fastener forces induce compression in the skin.

Output file (cont.):

Program REMACHADO. Example 1.
Crack with central stiffener intact.
Material: Skin: 2024 T3. Stiffeners: 7075 T73511.
Uniform fastening. 15 fasteners per stringer.

PAGE 21

```
*****
*
*   <<< F O R C E S   O N   F A S T E N E R S >>>
*
*****
```

Half crack length: 140.000

FASTENER	X = 0.00	X = 100.00	X = 200.00
-----	-----	-----	-----
1	-1851.3	-3046.1	1.1
2	-1603.3	-3112.4	131.0
3	-1599.2	-3070.3	460.4
4	-1726.8	-2854.0	841.2
5	-1909.8	-2602.0	1157.3
6	-2101.0	-2392.2	1376.7
7	-2281.4	-2251.5	1510.7
8	-2453.6	-2185.6	1583.5
9	-2634.7	-2195.4	1620.4
10	-2852.4	-2285.7	1644.3
11	-3143.7	-2469.4	1676.9
12	-3558.8	-2772.9	1741.7
13	-4171.3	-3244.2	1869.1
14	-5106.3	-3974.7	2106.3
15	-6632.0	-5169.2	2548.2

Positive fastener forces induce compression in the skin.

Output file (cont.):

Program REMACHADO. Example 1.
Crack with central stiffener intact.
Material: Skin: 2024 T3. Stiffeners: 7075 T73511.
Uniform fastening. 15 fasteners per stringer.

PAGE 22

```
*****
*
*   <<< F O R C E S   O N   F A S T E N E R S >>>
*
*****
```

Half crack length: 150.000

FASTENER	X = 0.00	X = 100.00	X = 200.00
-----	-----	-----	-----
1	-1836.2	-2683.0	5.9
2	-1568.0	-2702.1	219.5
3	-1541.8	-2764.5	671.9
4	-1650.7	-2696.5	1120.3
5	-1824.9	-2558.7	1447.1
6	-2019.0	-2421.2	1647.1
7	-2211.9	-2322.2	1751.8
8	-2403.0	-2279.2	1795.8
9	-2607.0	-2302.1	1808.4
10	-2849.8	-2401.4	1814.1
11	-3168.1	-2594.5	1835.1
12	-3613.1	-2911.0	1895.4
13	-4260.9	-3403.0	2026.6
14	-5240.8	-4167.2	2278.5
15	-6830.6	-5419.1	2752.9

Positive fastener forces induce compression in the skin.

Output file (cont.):

Program REMACHADO. Example 1.
Crack with central stiffener intact.
Material: Skin: 2024 T3. Stiffeners: 7075 T73511.
Uniform fastening. 15 fasteners per stringer.

PAGE 23

```
*****
*
*   <<< F O R C E S   O N   F A S T E N E R S >>>
*
*****
```

Half crack length: 160.000

FASTENER	X = 0.00	X = 100.00	X = 200.00
-----	-----	-----	-----
1	-1811.6	-2440.1	23.3
2	-1548.5	-2406.2	392.6
3	-1501.2	-2500.6	1000.8
4	-1591.0	-2524.6	1494.3
5	-1754.1	-2478.3	1797.9
6	-1946.9	-2409.4	1951.9
7	-2147.7	-2356.2	2009.7
8	-2353.5	-2342.4	2013.9
9	-2576.7	-2384.2	1995.9
10	-2841.7	-2497.3	1980.0
11	-3184.8	-2703.0	1987.7
12	-3657.8	-3034.9	2042.8
13	-4338.8	-3548.6	2177.4
14	-5360.9	-4346.2	2443.7
15	-7010.7	-5653.7	2950.0

Positive fastener forces induce compression in the skin.

Output file (cont.):

Program REMACHADO. Example 1.
Crack with central stiffener intact.
Material: Skin: 2024 T3. Stiffeners: 7075 T73511.
Uniform fastening. 15 fasteners per stringer.

PAGE 24

```
*****
*
*   <<< F O R C E S   O N   F A S T E N E R S >>>
*
*****
```

Half crack length: 170.000

FASTENER	X = 0.00	X = 100.00	X = 200.00
-----	-----	-----	-----
1	-1819.5	-2273.0	75.5
2	-1527.8	-2193.4	734.5
3	-1465.5	-2285.2	1500.8
4	-1539.7	-2359.8	1978.2
5	-1692.1	-2380.5	2207.4
6	-1882.1	-2370.9	2283.3
7	-2087.9	-2362.5	2276.0
8	-2305.2	-2380.3	2230.7
9	-2544.6	-2444.4	2177.3
10	-2829.1	-2574.4	2137.6
11	-3194.7	-2795.5	2131.3
12	-3693.6	-3144.2	2181.1
13	-4405.3	-3680.1	2319.3
14	-5466.6	-4510.5	2600.3
15	-7171.9	-5871.0	3138.0

Positive fastener forces induce compression in the skin.

Output file (cont.):

Program REMACHADO. Example 1.
Crack with central stiffener intact.
Material: Skin: 2024 T3. Stiffeners: 7075 T73511.
Uniform fastening. 15 fasteners per stringer.

PAGE 25

```
*****
*
*   <<< F O R C E S   O N   F A S T E N E R S >>>
*
*****
```

Half crack length: 180.000

FASTENER	X = 0.00	X = 100.00	X = 200.00
-----	-----	-----	-----
1	-1818.7	-2157.1	238.0
2	-1517.1	-2041.1	1404.9
3	-1440.9	-2115.8	2223.0
4	-1500.3	-2213.8	2570.0
5	-1641.7	-2279.8	2659.8
6	-1827.0	-2317.7	2625.1
7	-2035.1	-2349.4	2537.4
8	-2260.8	-2398.3	2436.1
9	-2513.2	-2485.8	2345.0
10	-2814.5	-2634.4	2281.4
11	-3199.9	-2871.9	2261.8
12	-3722.0	-3238.2	2307.2
13	-4461.6	-3796.2	2450.0
14	-5558.6	-4658.0	2745.9
15	-7314.2	-6068.1	3314.6

Positive fastener forces induce compression in the skin.

Output file (cont.):

Program REMACHADO. Example 1.
Crack with central stiffener intact.
Material: Skin: 2024 T3. Stiffeners: 7075 T73511.
Uniform fastening. 15 fasteners per stringer.

PAGE 26

```
*****
*
*   <<< F O R C E S   O N   F A S T E N E R S >>>
*
*****
```

Half crack length: 190.000

FASTENER	X = 0.00	X = 100.00	X = 200.00
-----	-----	-----	-----
1	-1815.8	-2078.1	870.5
2	-1512.6	-1934.2	2628.2
3	-1423.8	-1987.6	3169.0
4	-1470.2	-2092.4	3231.5
5	-1601.3	-2186.5	3119.9
6	-1781.3	-2259.8	2950.9
7	-1989.9	-2325.0	2775.1
8	-2221.5	-2401.8	2616.8
9	-2484.3	-2511.6	2489.6
10	-2799.3	-2678.5	2404.5
11	-3201.6	-2932.6	2374.0
12	-3743.8	-3316.1	2417.1
13	-4507.9	-3895.0	2565.8
14	-5636.3	-4785.8	2877.3
15	-7436.1	-6240.9	3476.5

Positive fastener forces induce compression in the skin.

Output file (cont.):

PAGE 27

Program REMACHADO. Example 1.
Crack with central stiffener intact.
Material: Skin: 2024 T3. Stiffeners: 7075 T73511.
Uniform fastening. 15 fasteners per stringer.

```
*****
*
*   <<< F O R C E S   O N   F A S T E N E R S >>>
*
*****
```

Half crack length: 200.000

FASTENER	X = 0.00	X = 100.00	X = 200.00
-----	-----	-----	-----
1	-1818.4	-2037.1	-13514.4
2	-1461.3	-1812.2	-7173.0
3	-1335.9	-1821.8	-4251.8
4	-1355.8	-1920.3	-2655.5
5	-1473.0	-2031.1	-1743.2
6	-1652.2	-2129.8	-1225.7
7	-1871.7	-2221.9	-952.0
8	-2124.0	-2325.7	-838.4
9	-2416.0	-2464.8	-837.9
10	-2768.1	-2666.6	-926.7
11	-3215.5	-2965.5	-1096.8
12	-3813.0	-3408.4	-1354.7
13	-4647.1	-4067.2	-1723.7
14	-5868.8	-5067.9	-2255.8
15	-7804.9	-6685.4	-3076.1

Positive fastener forces induce compression in the skin.

Output file (cont.):

PAGE 28

Program REMACHADO. Example 1.
Crack with central stiffener intact.
Material: Skin: 2024 T3. Stiffeners: 7075 T73511.
Variable fastening. 5 fasteners per stringer.

```
*****
*                                     *
*   <<< I N P U T   D A T A >>>   *
*                                     *
*****
```

EXECUTION DATA:

=====

Date: 16-Mar-2018
Remachado 3.2. (c) Airbus Defence and Space. Mar-2018

SKIN DATA:

=====

Young's modulus.....:	72400.00
Poisson's ratio.....:	0.33
Thickness.....:	2.00
Fracture toughness Kc.....:	5000.00
Transversal supports spacing..:	500.00

APPLIED STRESSES:

=====

Average axial stress.....:	200.00
Lateral-axial stress ratio...:	0.50

STIFFENERS DATA:

=====

	STIFFENER 1	STIFFENER 2	STIFFENER 3
	-----	-----	-----
X-coordinate.....:	0.00	100.00	200.00
No uniform fastener system:			
Number of fasteners.....:	5	5	5
Thickness at attach point.:	3.00	3.00	3.00
Gross area.....:	237.00	237.00	237.00
Net area.....:	222.60	222.60	222.60
Gross moment of inertia.....:	44610.00	44610.00	44610.00
Net moment of inertia.....:	40674.00	40674.00	40674.00
Distance from CG to outer cap:	17.50	17.50	17.50
Distance from CG to inner cap:	17.50	17.50	17.50
Young's modulus.....:	71700.00	71700.00	71700.00
Ftu of stiffener.....:	469.00	469.00	469.00
Fty of stiffener.....:	400.00	400.00	400.00
Initial state of stiffener...:	Intact	Intact	Intact

Output file (cont.):

PAGE 29

Program REMACHADO. Example 1.
Crack with central stiffener intact.
Material: Skin: 2024 T3. Stiffeners: 7075 T73511.
Variable fastening. 5 fasteners per stringer.

```
*****
*                                     *
*   <<< I N P U T   D A T A >>>   *
*                                     *
*****
```

FASTENERS FOR STIFFENER AT X = 0.00:

=====

NUMBER	DIAMETER	X	Y	MODULUS	STIFFNESS
-----	-----	-----	-----	-----	-----
1	4.80	0.00	10.00	71000.00	33865.65
2	4.80	0.00	30.00	71000.00	33865.65
3	4.80	0.00	50.00	71000.00	33865.65
4	4.80	0.00	70.00	71000.00	33865.65
5	4.80	0.00	90.00	71000.00	33865.65

FASTENERS FOR STIFFENER AT X = 100.00:

=====

NUMBER	DIAMETER	X	Y	MODULUS	STIFFNESS
-----	-----	-----	-----	-----	-----
1	4.80	100.00	10.00	71000.00	33865.65
2	4.80	100.00	30.00	71000.00	33865.65
3	4.80	100.00	50.00	71000.00	33865.65
4	4.80	100.00	70.00	71000.00	33865.65
5	4.80	100.00	90.00	71000.00	33865.65

FASTENERS FOR STIFFENER AT X = 200.00:

=====

NUMBER	DIAMETER	X	Y	MODULUS	STIFFNESS
-----	-----	-----	-----	-----	-----
1	4.80	200.00	10.00	71000.00	33865.65
2	4.80	200.00	30.00	71000.00	33865.65
3	4.80	200.00	50.00	71000.00	33865.65
4	4.80	200.00	70.00	71000.00	33865.65
5	4.80	200.00	90.00	71000.00	33865.65

Output file (cont.):

PAGE 30

Program REMACHADO. Example 1.
Crack with central stiffener intact.
Material: Skin: 2024 T3. Stiffeners: 7075 T73511.
Variable fastening. 5 fasteners per stringer.

```
*****
*                                     *
*   <<< R E S U L T S   F O R   S K I N >>>   *
*                                     *
*****
```

APPLIED STRESSES:

=====

Average axial stress (YD).....: 200.00
Ratio of stresses (XD/YD).....: 0.50
Unit axial skin stress.....: 218.87
Unit lateral skin stress.....: 100.00

HALF CRACK LENGTH	STRESS IN THE FAR FIELD	BETA FACTOR	STRESS INTENSITY FACTOR	STIFFENERS STATUS
-----	-----	-----	-----	-----
0.00	218.87	1.00000	0.00	III
10.00	218.87	0.92521	1135.03	III
20.00	218.87	0.88230	1530.71	III
30.00	218.87	0.85798	1823.07	III
40.00	218.87	0.84130	2064.17	III
50.00	218.87	0.82862	2273.02	III
60.00	218.87	0.81743	2456.35	III
70.00	218.87	0.80535	2613.96	III
80.00	218.87	0.78903	2737.82	III
90.00	218.87	0.76133	2801.94	III
100.00	218.87	0.67965	2636.64	III
110.00	218.87	0.59176	2407.72	III
120.00	218.87	0.57747	2454.05	III
130.00	218.87	1.85375	8199.51	XXI
140.00	218.87	1.77278	8137.34	XXI
150.00	218.87	1.69933	8073.98	XXI
160.00	218.87	1.62860	7991.69	XXI
170.00	218.87	1.55589	7869.86	XXI
180.00	218.87	1.47457	7674.78	XXI
190.00	218.87	1.37103	7331.40	XXI
200.00	218.87	2.15317	11812.94	XXX

Beta factors referred to the unit axial skin stress.

Key for stiffeners status: I=Intact
X=Failed

6.3 Example 2

In this example, the previous problem will be solved with the following additional considerations, in the two runs of the program:

- Run 1:

Fifteen fasteners (at each side of the crack) per stiffener will be considered for the load transfer between the stiffeners and the skin.

In the input data, a uniform fastening will be specified with the data defined in the previous section.

The program, by means of the expression given in section 3.2, will automatically calculate the stiffness of the fasteners. For this purpose, a null stiffness will be specified in the input data.

The first stiffener will be considered initially broken and that the rest intact. The program will be allowed to determine the breakdown of the second, but it will not be allowed the failure of the third one.

It will be requested results for the skin, for the stiffeners and allowable stresses.

- Run 2:

Five fasteners (at each side of the crack) per stiffener will be considered for the load transfer between the stiffeners and the skin.

In the input data, a variable fastening will be specified with the data defined previously, and so, in reality, it will be in the same way as in the first step.

The stiffness of the fasteners will be specified in the data input, as previously defined, which coincides with the one calculated by the program in the first step.

The state of the stiffeners will be considered just like for the first run (The first is broken and the other two are intact).

Results will only be requested for the skin.

The difference between both runs lies in the definition of the fastening, although it will formally be the same in both cases, except for the number of fasteners.

Therefore, this example could serve to evaluate the influence of the total number of fasteners considered per stringer.

With this data, we can prepare the input file. This file as well as the output file obtained by running the program is shown on next pages:

Input file:

```

1 Program REMACHADO. Example 2.
  Crack with central stiffener failed. Stiffener 3 never fails.
  Material: Skin: 2024 T3. Stiffeners: 7075 T73511.
  Uniform fastening. 15 fasteners per stringer.
2 SKIN DATA: E  μ  t  Kc  L
  72400.0      0.33      2.0      5000.0      500.0
3 STRINGERS DATA: nl  X  ts  Ag  An  Ig  In  Zo  Zi  Es  FtU  Fty  Fail
  3      0.0      100.0      200.0
      3.0      3.0      3.0
      237.0      237.0      237.0
      222.6      222.6      222.6
      44610.0      44610.0      44610.0
      40674.0      40674.0      40674.0
      17.5      17.5      17.5
      17.5      17.5      17.5
      71700.0      71700.0      71700.0
      469.0      469.0      469.0
      400.0      400.0      400.0
      1      0      -1
4 UNIFORM FASTENING: nr  Pitch  D  Er  Sr
  15      20.00      4.80      71000.00      0.00
  15      20.00      4.80      71000.00      0.00
  15      20.00      4.80      71000.00      0.00
5 FAR FIELD STRESS: Reference_stress  Sx/Sy
  200.00      0.50
6 CRACK LENGTHS
  21  0.0  10.0  20.0  30.0  40.0  50.0
      60.0  70.0  80.0  90.0  100.0
      110.0  120.0  130.0  140.0  150.0
      160.0  170.0  180.0  190.0  200.0
7 PRINT CONTROL: Skin  Stringers  Allowables  Forces
  1  1  1  0
8 RUN THE PROGRAM
1 Program REMACHADO. Example 2.
  Crack with central stiffener failed. Stiffener 3 never fails.
  Material: Skin: 2024 T3. Stiffeners: 7075 T73511.
  Variable fastening. 5 fasteners per stringer.
4 VARIABLE FASTENING: n  X(i)  Y(i)  D(i)  E(i)  K(i)
  5      0.00      10.00      4.80      71000.00      33865.65
      0.00      30.00      4.80      71000.00      33865.65
      0.00      50.00      4.80      71000.00      33865.65
      0.00      70.00      4.80      71000.00      33865.65
      0.00      90.00      4.80      71000.00      33865.65
  5      100.00      10.00      4.80      71000.00      33865.65
      100.00      30.00      4.80      71000.00      33865.65
      100.00      50.00      4.80      71000.00      33865.65
      100.00      70.00      4.80      71000.00      33865.65
      100.00      90.00      4.80      71000.00      33865.65
  5      200.00      10.00      4.80      71000.00      33865.65
      200.00      30.00      4.80      71000.00      33865.65
      200.00      50.00      4.80      71000.00      33865.65
      200.00      70.00      4.80      71000.00      33865.65
      200.00      90.00      4.80      71000.00      33865.65
7 PRINT CONTROL: Skin  Stringers  Allowables  Forces
  1  0  0  0
8 RUN THE PROGRAM

```

Output file:

Program REMACHADO. Example 2.
Crack with central stiffener failed. Stiffener 3 never fails.
Material: Skin: 2024 T3. Stiffeners: 7075 T73511.
Uniform fastening. 15 fasteners per stringer.

PAGE 1

```
*****
*                                     *
*   <<< I N P U T   D A T A >>>   *
*                                     *
*****
```

EXECUTION DATA:

=====

Date: 16-Mar-2018

Remachado 3.2. (c) Airbus Defence and Space. Mar-2018

SKIN DATA:

=====

Young's modulus.....:	72400.00
Poisson's ratio.....:	0.33
Thickness.....:	2.00
Fracture toughness Kc.....:	5000.00
Transversal supports spacing..:	500.00

APPLIED STRESSES:

=====

Average axial stress.....:	200.00
Lateral-axial stress ratio...:	0.50

STIFFENERS DATA:

=====

	STIFFENER 1	STIFFENER 2	STIFFENER 3
	-----	-----	-----
X-coordinate.....:	0.00	100.00	200.00
Uniform fastener system:			
Number of fasteners.....:	15	15	15
Fastener pitch.....:	20.00	20.00	20.00
Fastener Young's modulus..:	71000.00	71000.00	71000.00
Fastener stiffness.....:	0.00	0.00	0.00
Fastener diameter.....:	4.80	4.80	4.80
Thickness at attach point..:	3.00	3.00	3.00
Gross area.....:	237.00	237.00	237.00
Net area.....:	222.60	222.60	222.60
Gross moment of inertia.....:	44610.00	44610.00	44610.00
Net moment of inertia.....:	40674.00	40674.00	40674.00
Distance from CG to outer cap:	17.50	17.50	17.50
Distance from CG to inner cap:	17.50	17.50	17.50
Young's modulus.....:	71700.00	71700.00	71700.00
Ftu of stiffener.....:	469.00	469.00	469.00
Fty of stiffener.....:	400.00	400.00	400.00
Initial state of stiffener....:	Failed	Intact	No failure

Output file (cont.):

Program REMACHADO. Example 2.
Crack with central stiffener failed. Stiffener 3 never fails.
Material: Skin: 2024 T3. Stiffeners: 7075 T73511.
Uniform fastening. 15 fasteners per stringer.

PAGE 2

```
*****
*                                     *
*   <<< R E S U L T S   F O R   S K I N >>>   *
*                                     *
*****
```

APPLIED STRESSES:

=====

Average axial stress (YD).....: 200.00
Ratio of stresses (XD/YD).....: 0.50
Unit axial skin stress.....: 218.87
Unit lateral skin stress.....: 100.00

HALF CRACK LENGTH	STRESS IN THE FAR FIELD	BETA FACTOR	STRESS INTENSITY FACTOR	STIFFENERS STATUS
-----	-----	-----	-----	-----
0.00	218.87	2.32487	0.00	XII
10.00	218.87	1.86966	2293.65	XII
20.00	218.87	1.59272	2763.24	XII
30.00	218.87	1.45849	3099.05	XII
40.00	218.87	1.37206	3366.40	XII
50.00	218.87	1.30694	3585.15	XII
60.00	218.87	1.25185	3761.79	XII
70.00	218.87	1.19995	3894.73	XII
80.00	218.87	1.14491	3972.67	XII
90.00	218.87	1.07596	3959.86	XII
100.00	218.87	0.93342	3621.11	XII
110.00	218.87	0.78556	3196.23	XII
120.00	218.87	0.74288	3156.99	XII
130.00	218.87	1.57790	6979.36	XXI
140.00	218.87	1.51656	6961.27	XXI
150.00	218.87	1.46231	6947.83	XXI
160.00	218.87	1.41043	6921.10	XXI
170.00	218.87	1.35693	6863.52	XXI
180.00	218.87	1.29654	6748.20	XXI
190.00	218.87	1.21795	6512.85	XXI
200.00	218.87	1.05337	5779.12	XXI

Beta factors referred to the unit axial skin stress.

Key for stiffeners status: I=Intact
X=Failed

Output file (cont.):

Program REMACHADO. Example 2.

PAGE 3

Crack with central stiffener failed. Stiffener 3 never fails.

Material: Skin: 2024 T3. Stiffeners: 7075 T73511.

Uniform fastening. 15 fasteners per stringer.

*
* <<< R E S U L T S F O R S T I F F E N E R A T X = 0.00 >>> *
*

Unit stiffener stress: 184.07

HALF CRACK LENGTH	AXIAL FORCE	BENDING MOMENT	GROSS STRESS		NET STRESS	
			-----		-----	
			OUTER CAP	INNER CAP	OUTER CAP	INNER CAP
-----	-----	-----	-----	-----	-----	-----
0.00	0.0	0.0	0.00	0.00	0.00	0.00
10.00	0.0	0.0	0.00	0.00	0.00	0.00
20.00	0.0	0.0	0.00	0.00	0.00	0.00
30.00	0.0	0.0	0.00	0.00	0.00	0.00
40.00	0.0	0.0	0.00	0.00	0.00	0.00
50.00	0.0	0.0	0.00	0.00	0.00	0.00
60.00	0.0	0.0	0.00	0.00	0.00	0.00
70.00	0.0	0.0	0.00	0.00	0.00	0.00
80.00	0.0	0.0	0.00	0.00	0.00	0.00
90.00	0.0	0.0	0.00	0.00	0.00	0.00
100.00	0.0	0.0	0.00	0.00	0.00	0.00
110.00	0.0	0.0	0.00	0.00	0.00	0.00
120.00	0.0	0.0	0.00	0.00	0.00	0.00
130.00	0.0	0.0	0.00	0.00	0.00	0.00
140.00	0.0	0.0	0.00	0.00	0.00	0.00
150.00	0.0	0.0	0.00	0.00	0.00	0.00
160.00	0.0	0.0	0.00	0.00	0.00	0.00
170.00	0.0	0.0	0.00	0.00	0.00	0.00
180.00	0.0	0.0	0.00	0.00	0.00	0.00
190.00	0.0	0.0	0.00	0.00	0.00	0.00
200.00	0.0	0.0	0.00	0.00	0.00	0.00

Loads and stresses for stiffener at Y = 0.

Output file (cont.):

Program REMACHADO. Example 2.

PAGE 4

Crack with central stiffener failed. Stiffener 3 never fails.

Material: Skin: 2024 T3. Stiffeners: 7075 T73511.

Uniform fastening. 15 fasteners per stringer.

```
*****
*
*   <<< R E S U L T S   F O R   S T I F F E N E R   A T   X = 100.00 >>>
*
*****
```

Unit stiffener stress: 184.07

HALF CRACK LENGTH	AXIAL FORCE	BENDING MOMENT	GROSS STRESS		NET STRESS	
			OUTER CAP	INNER CAP	OUTER CAP	INNER CAP
0.00	46206.2	5881.5	197.27	192.66	210.11	205.04
10.00	46477.9	6944.4	198.83	193.39	211.78	205.81
20.00	47067.4	9395.3	202.28	194.91	215.49	207.40
30.00	47895.9	13125.3	207.24	196.94	220.81	209.52
40.00	48969.1	18453.7	213.86	199.38	227.93	212.05
50.00	50312.6	25944.7	222.47	202.11	237.19	214.86
60.00	51971.1	36489.7	233.60	204.97	249.17	217.77
70.00	54017.9	51517.1	248.13	207.71	264.83	220.50
80.00	56576.8	73461.9	267.54	209.90	285.77	222.56
90.00	59913.5	107536.9	294.98	210.61	315.42	222.89
100.00	64801.1	169107.7	339.76	207.08	363.87	218.35
110.00	70113.6	242892.1	391.12	200.55	419.48	210.47
120.00	74265.5	298560.4	430.48	196.23	462.08	205.17
130.00	0.0	0.0	0.00	0.00	0.00	0.00
140.00	0.0	0.0	0.00	0.00	0.00	0.00
150.00	0.0	0.0	0.00	0.00	0.00	0.00
160.00	0.0	0.0	0.00	0.00	0.00	0.00
170.00	0.0	0.0	0.00	0.00	0.00	0.00
180.00	0.0	0.0	0.00	0.00	0.00	0.00
190.00	0.0	0.0	0.00	0.00	0.00	0.00
200.00	0.0	0.0	0.00	0.00	0.00	0.00

Loads and stresses for stiffener at Y = 0.

Output file (cont.):

Program REMACHADO. Example 2.

PAGE 5

Crack with central stiffener failed. Stiffener 3 never fails.

Material: Skin: 2024 T3. Stiffeners: 7075 T73511.

Uniform fastening. 15 fasteners per stringer.

```
*****
*
*   <<< R E S U L T S   F O R   S T I F F E N E R   A T   X = 200.00 >>>
*
*****
```

Unit stiffener stress: 184.07

HALF CRACK LENGTH	AXIAL FORCE	BENDING MOMENT	GROSS STRESS		NET STRESS	
			OUTER CAP	INNER CAP	OUTER CAP	INNER CAP
0.00	44505.1	303.4	187.90	187.67	200.06	199.80
10.00	44584.8	289.9	188.24	188.01	200.42	200.17
20.00	44756.3	269.8	188.95	188.74	201.18	200.95
30.00	44992.8	255.0	189.94	189.74	202.23	202.01
40.00	45290.1	261.0	191.20	191.00	203.57	203.35
50.00	45646.7	314.7	192.73	192.48	205.20	204.93
60.00	46061.9	455.8	194.53	194.18	207.12	206.73
70.00	46535.0	737.7	196.64	196.06	209.37	208.73
80.00	47063.2	1228.3	199.06	198.10	211.95	210.90
90.00	47637.2	2004.7	201.79	200.21	214.87	213.14
100.00	48215.4	3098.3	204.66	202.23	217.93	215.27
110.00	48703.4	4363.2	207.21	203.79	220.67	216.92
120.00	49196.0	6033.0	209.94	205.21	223.60	218.41
130.00	61388.7	31891.6	271.53	246.51	289.50	262.06
140.00	63894.4	41743.1	285.97	253.22	305.00	269.08
150.00	66696.2	55051.5	303.01	259.82	323.31	275.94
160.00	69887.6	73287.6	323.63	266.13	345.49	282.43
170.00	73597.0	98709.2	349.26	271.81	373.09	288.15
180.00	78025.8	135100.1	382.22	276.22	408.65	292.39
190.00	83592.3	190558.2	427.46	277.96	457.51	293.54
200.00	91498.3	288964.2	499.43	272.71	535.37	286.72

Loads and stresses for stiffener at Y = 0.

Output file (cont.):

Program REMACHADO. Example 2.

PAGE 6

Crack with central stiffener failed. Stiffener 3 never fails.

Material: Skin: 2024 T3. Stiffeners: 7075 T73511.

Uniform fastening. 15 fasteners per stringer.

```
*****
*                                     *
*   <<< A L L O W E D   S T R E S S E S >>>   *
*                                     *
*****
```

HALF CRACK LENGTH	SKIN	STIFFENER 1		STIFFENER 2		STIFFENER 3	
		Smax=Ftu	Smed=Fty	Smax=Ftu	Smed=Fty	Smax=Ftu	Smed=Fty
0.00	*****	0.00	0.00	446.44	385.40	468.85	400.13
10.00	435.99	0.00	0.00	442.91	383.15	468.03	399.42
20.00	361.89	0.00	0.00	435.30	378.35	466.25	397.89
30.00	322.68	0.00	0.00	424.79	371.81	463.82	395.80
40.00	297.05	0.00	0.00	411.54	363.66	460.77	393.20
50.00	278.93	0.00	0.00	395.47	353.95	457.12	390.13
60.00	265.83	0.00	0.00	376.45	342.65	452.87	386.61
70.00	256.76	0.00	0.00	354.19	329.67	448.01	382.68
80.00	251.72	0.00	0.00	328.24	314.76	442.55	378.38
90.00	252.53	0.00	0.00	297.38	297.23	436.55	373.83
100.00	276.16	0.00	0.00	257.79	274.81	430.41	369.34
110.00	312.87	0.00	0.00	223.61	253.99	425.07	365.64
120.00	316.76	0.00	0.00	202.99	239.79	419.50	361.98
130.00	143.28	0.00	0.00	0.00	0.00	324.00	290.09
140.00	143.65	0.00	0.00	0.00	0.00	307.54	278.71
150.00	143.93	0.00	0.00	0.00	0.00	290.12	267.00
160.00	144.49	0.00	0.00	0.00	0.00	271.50	254.81
170.00	145.70	0.00	0.00	0.00	0.00	251.41	241.97
180.00	148.19	0.00	0.00	0.00	0.00	229.54	228.23
190.00	153.54	0.00	0.00	0.00	0.00	205.02	213.03
200.00	173.04	0.00	0.00	0.00	0.00	175.21	194.63

Allowed stresses are based on the reference stress.

Output file (cont.):

Program REMACHADO. Example 2.
Crack with central stiffener failed. Stiffener 3 never fails.
Material: Skin: 2024 T3. Stiffeners: 7075 T73511.
Variable fastening. 5 fasteners per stringer.

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*****
*                                     *
*   <<< I N P U T   D A T A >>>   *
*                                     *
*****
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EXECUTION DATA:

=====

Date: 16-Mar-2018

Remachado 3.2. (c) Airbus Defence and Space. Mar-2018

SKIN DATA:

=====

Young's modulus.....:	72400.00
Poisson's ratio.....:	0.33
Thickness.....:	2.00
Fracture toughness Kc.....:	5000.00
Transversal supports spacing...:	500.00

APPLIED STRESSES:

=====

Average axial stress.....:	200.00
Lateral-axial stress ratio...:	0.50

STIFFENERS DATA:

=====

	STIFFENER 1	STIFFENER 2	STIFFENER 3
	-----	-----	-----
X-coordinate.....:	0.00	100.00	200.00
No uniform fastener system:			
Number of fasteners.....:	5	5	5
Thickness at attach point.:	3.00	3.00	3.00
Gross area.....:	237.00	237.00	237.00
Net area.....:	222.60	222.60	222.60
Gross moment of inertia.....:	44610.00	44610.00	44610.00
Net moment of inertia.....:	40674.00	40674.00	40674.00
Distance from CG to outer cap:	17.50	17.50	17.50
Distance from CG to inner cap:	17.50	17.50	17.50
Young's modulus.....:	71700.00	71700.00	71700.00
Ftu of stiffener.....:	469.00	469.00	469.00
Fty of stiffener.....:	400.00	400.00	400.00
Initial state of stiffener...:	Failed	Intact	No failure

Output file (cont.):

Program REMACHADO. Example 2.
Crack with central stiffener failed. Stiffener 3 never fails.
Material: Skin: 2024 T3. Stiffeners: 7075 T73511.
Variable fastening. 5 fasteners per stringer.

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*****
*                                     *
*   <<< I N P U T   D A T A >>>   *
*                                     *
*****
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FASTENERS FOR STIFFENER AT X = 0.00:

=====

NUMBER	DIAMETER	X	Y	MODULUS	STIFFNESS
-----	-----	-----	-----	-----	-----
1	4.80	0.00	10.00	71000.00	33865.65
2	4.80	0.00	30.00	71000.00	33865.65
3	4.80	0.00	50.00	71000.00	33865.65
4	4.80	0.00	70.00	71000.00	33865.65
5	4.80	0.00	90.00	71000.00	33865.65

FASTENERS FOR STIFFENER AT X = 100.00:

=====

NUMBER	DIAMETER	X	Y	MODULUS	STIFFNESS
-----	-----	-----	-----	-----	-----
1	4.80	100.00	10.00	71000.00	33865.65
2	4.80	100.00	30.00	71000.00	33865.65
3	4.80	100.00	50.00	71000.00	33865.65
4	4.80	100.00	70.00	71000.00	33865.65
5	4.80	100.00	90.00	71000.00	33865.65

FASTENERS FOR STIFFENER AT X = 200.00:

=====

NUMBER	DIAMETER	X	Y	MODULUS	STIFFNESS
-----	-----	-----	-----	-----	-----
1	4.80	200.00	10.00	71000.00	33865.65
2	4.80	200.00	30.00	71000.00	33865.65
3	4.80	200.00	50.00	71000.00	33865.65
4	4.80	200.00	70.00	71000.00	33865.65
5	4.80	200.00	90.00	71000.00	33865.65

Output file (cont.):

Program REMACHADO. Example 2.
Crack with central stiffener failed. Stiffener 3 never fails.
Material: Skin: 2024 T3. Stiffeners: 7075 T73511.
Variable fastening. 5 fasteners per stringer.

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*****
*                                     *
*   <<< R E S U L T S   F O R   S K I N >>>   *
*                                     *
*****
```

APPLIED STRESSES:

=====

Average axial stress (YD).....: 200.00
Ratio of stresses (XD/YD).....: 0.50
Unit axial skin stress.....: 218.87
Unit lateral skin stress.....: 100.00

HALF CRACK LENGTH	STRESS IN THE FAR FIELD	BETA FACTOR	STRESS INTENSITY FACTOR	STIFFENERS STATUS
-----	-----	-----	-----	-----
0.00	218.87	3.05816	0.00	XII
10.00	218.87	2.39205	2934.50	XII
20.00	218.87	1.97758	3430.95	XII
30.00	218.87	1.77173	3764.64	XII
40.00	218.87	1.63686	4016.10	XII
50.00	218.87	1.53529	4211.54	XII
60.00	218.87	1.45117	4360.73	XII
70.00	218.87	1.37483	4462.33	XII
80.00	218.87	1.29747	4502.00	XII
90.00	218.87	1.20549	4436.59	XII
100.00	218.87	1.03106	3999.89	XII
110.00	218.87	0.85664	3485.46	XII
120.00	218.87	0.80398	3416.64	XII
130.00	218.87	1.85375	8199.51	XXI
140.00	218.87	1.77278	8137.34	XXI
150.00	218.87	1.69933	8073.98	XXI
160.00	218.87	1.62860	7991.69	XXI
170.00	218.87	1.55589	7869.86	XXI
180.00	218.87	1.47457	7674.78	XXI
190.00	218.87	1.37103	7331.40	XXI
200.00	218.87	1.16859	6411.24	XXI

Beta factors referred to the unit axial skin stress.

Key for stiffeners status: I=Intact
X=Failed

